Factor-Augmented Black–Litterman Portfolio Construction with Higher-Moment Optimisation and Covariance Shrinkage

A Multi-Estimator Empirical Study on a Diversified ETF Universe

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Abstract

In this paper, we develop a factor-augmented Black–Litterman portfolio framework that integrates Bayesian return estimation, multi-estimator covariance analysis, higher-moment optimisation, and risk-parity diversification. We benchmark five covariance estimators across three risk-aversion regimes and four optimisation objectives using twelve ETFs spanning nine asset classes over 1,509 daily observations. A threshold condition number at $\kappa \approx 10$ separates estimators into different regimes: the Ledoit–Wolf estimator activates all twelve assets, while four ill-conditioned alternatives converge to concentrated six-asset portfolios. The BL-Optimal portfolio generates a strong annualised alpha of +17.15% ($t = 2.08$, $p = 0.037$), which is confirmed against the equal-weight benchmark by the Jobson–Korkie test ($z = +2.42$, $p = 0.016$). The second-order stochastic dominance comparison shows that the higher moment portfolio dominates the minimum CVaR allocation. This research is the first multi-estimator covariance propagation analysis in a full BL pipeline, and the first integration of BL-posterior returns with MVSK optimisation.

Keywords Black–Litterman, covariance shrinkage, Fama–French factors, higher-moment, risk parity, CVaR, risk contribution, diversification, random matrix theory, Bayesian allocation

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1. Introduction

The construction of optimal multi-asset portfolios is one of the most important problems in quantitative finance. This problem is based on a fundamental tension: the theoretical elegance of mean-variance optimisation (Markowitz, 1952) is systematically undermined by parameter estimation error. Unconstrained MVO is an error maximiser as shown by Michaud (1989) concentrating weight in assets with over-estimated returns and underestimated variances leading to extreme, unstable and economically unintuitive allocations. The magnitude was measured by DeMiguel, Garlappi and Uppal (2009) who show that the naive 1/N portfolio often outperforms sophisticated optimised allocations when the estimation error is large relative to the true Sharpe ratio dispersion.

The most influential Bayesian response has been the Black–Litterman model (Black & Litterman, 1991, 1992). The Black–Litterman model anchors expected returns to a market-equilibrium prior obtained through CAPM reverse-optimisation ($\Pi = \delta\Sigma\tilde{w}$) and combines this with investor views through precision weighted posterior updating. Idzorek (2007) introduced a confidence calibration that transforms subjective conviction into Ω. Kolm and Ritter (2017) formalised the Bayesian interpretation with $\tau\Sigma$ parameterising prior uncertainty. He and Litterman (1999) provided geometric intuition. Huynh and Ta (2025) extended view generation using deep learning.

While individual maturity has advanced, the literature has developed Bayesian allocation, covariance shrinkage, factor-based pricing, higher-moment theory, risk-parity diversification in isolation, leaving five critical gaps (i) no analysis of how covariance estimator choice propagates through the four BL channels (Π, $\tau\Sigma$, Ω, Σ_BL) despite nonlinear error amplification across interdependent matrix inversions; (ii) fragmented single-stage factor integration risking tilt neutralisation during optimisation; (iii) disconnection between BL-posterior returns and higher-moment objectives (MVSK/CVaR); (iv) absent ML-driven feature engineering for BL view construction; (v) inadequate out-of-sample validation across structurally distinct regimes.

This study closes all five gaps through a seventeen-stage pipeline: multi-stage screening (correlation pruning at $|\rho| < 0.70$, Ward clustering, Newey–West factor-beta regression) reduces ninety-six ETFs to twelve covering nine asset classes. We benchmark five covariance estimators under three risk-aversion regimes and four objectives, using a seven-factor model (FF5, momentum, BAB). The contribution of the study is: (i) first multi-estimator covariance propagation analysis in a full BL pipeline, showing a $\kappa \approx 10$ condition number threshold separating estimators into different portfolio regimes; (ii) a tripartite factor approach providing end-to-end consistency; (iii) first BL-posterior integration with MVSK and CVaR objectives; (iv) an ML-driven feature pipeline for regime-aware view construction; and (v) regime-aware out-of-sample validation with strict temporal partitioning.

2. Research Objectives

Our central thesis is that allocation stability of the Black–Litterman (BL) posterior is fundamentally driven by covariance conditioning, not point-estimator accuracy, and that the economic value of the BL posterior is only fully realised when jointly conditioned on a well-posed second-moment operator and coupled to a higher-moment objective. The thesis has five objectives, each of which addresses a particular link in the inferential chain from raw returns to auditable out-of-sample performance. Ancillary tasks (rolling-beta descriptives, benchmark backtests, factor-timeseries diagnostics) are considered supporting instrumentation, and are subsumed into the objectives below, rather than elevated to first-order research questions.

Objective 1 – Development of a well-conditioned, factor-accessible investable universe. Truncate a broad universe of exchange-traded funds across at least five asset classes to a narrow set whose underlying rank is consistent with the empirically derived factor dimensionality. Admission is co-

governed by a greedy Pearson prune at $|\rho| < 0.70$, Ward-linkage clustering on the ultrametric distance $d = \sqrt{(2(1-\rho))}$, and a Marchenko–Pastur spectral cut reconciled against Shannon effective dimensionality. This objective implements the ex-ante numerical precondition, without which all subsequent inversions are unstable.

Objective 2 – Propagation of Covariance through BL Pipeline with Multiple Estimators. Benchmark five covariance estimators (Sample, Ledoit–Wolf linear and nonlinear, RMT-denoised, OAS) on condition number, Frobenius distance and spectral entropy, and - critically - propagate each estimator through all four BL channels ($\Pi = \delta\Sigma\tilde{w}$; $\tau\Sigma$; Idzorek Ω; $\Sigma_BL = [(\tau\Sigma)^{-1} + P'\Omega^{-1}P]^{-1}$) to terminal weights. This objective separates causal transmission from spectral conditioning to allocation dispersion and identifies the critical condition-number threshold that separates estimators into structurally different portfolio regimes. This is the paper's first-order methodological contribution.

Objective 3 – Bayesian Posterior Construction with Factor Informed Views. Derive the BL posterior in dual precision-weighted and Kalman-gain forms, verified to machine-precision numerical identity, under three risk-aversion calibrations (Trustee, Market-implied, Kelly) and a tau-grid sweep. Views are constructed on the seven-factor structure (Fama–French five plus UMD and de-leveraged BAB) using Idzorek confidence mapping that ensures end-to-end consistency between the factors that gate universe admission and those that discipline the posterior mean.

Objective 4 – Higher-Moment Optimisation with Inclusion of BL-Posterior Means. Feed μ_BL as the first moment input to an Athayde–Flores MVSK utility with coskewness (M_3) and cokurtosis (M_4) tensors, solved by SLSQP with fifty Dirichlet seeded restarts, and benchmark against constrained mean-variance, Dinkelbach maximum-Sharpe and Rockafellar–Uryasev minimum- $CVaR_5\%$. In particular, the objective is to test whether the Bayesian return regularisation and the distributional-asymmetry aversion, when imposed jointly, would produce allocations that stochastically dominate their single-preference counterparts, an integration that has been absent in the literature so far.

Objective 5 – Regime-aware out-of-sample validation and attribution. Evaluate optimised portfolios under strict temporal partitioning across structurally heterogeneous training and testing windows, with Sharpe differentials adjudicated by the Jobson–Korkie-Memmel test and autocorrelation-adjusted bootstrap intervals, and with first- and second-order stochastic dominance rankings imposed on a nineteen-metric scorecard. Decomposition of realised excess return via seven-factor attribution with Newey–West HAC errors and reconciliation with the Fundamental Law diagnostic (rolling information coefficient, effective breadth) to distinguish skill-driven alpha from mechanical factor replication.

3. Review of Literature

Portfolio construction under parameter uncertainty, non-Gaussian returns and time-varying factor structure is influenced by five research traditions that have developed relatively independently: Bayesian allocation, covariance estimation, cross-sectional pricing, higher-moment optimisation and risk-parity diversification. Here we review both foundational and modern contributions across six thematic domains, explicitly identifying where each literature informs, and where it does not, the integrative multi-estimator factor-augmented Black–Litterman framework developed here.

3.1 The Black–Litterman Model and Bayesian Portfolio Optimisation

The Black–Litterman (BL) model (Black & Litterman, 1991, 1992) is the canonical Bayesian cure for the instability of unconstrained mean-variance optimisation (Markowitz, 1952). It inverts the CAPM to imply an equilibrium prior $\Pi = \delta\Sigma\tilde{w}$ from market-capitalisation weights and combines this with investor views via a precision-weighted posterior update, substantially reducing the estimation-error sensitivity of the direct Markowitz solution. The prior is calibrated to market equilibrium rather than noisy sample

means, so the posterior degrades gracefully to the equilibrium allocation as view confidence tends to zero.

This is a continuity property lacking from classical MV. He and Litterman (1999) provide the economic geometry: the posterior lies on the line segment between the prior and the view-implied point, with its location determined by the confidence of the view. Satchell and Scowcroft (2000) break down BL weights into equilibrium and view components: $w^* = w_{eq} + \Sigma_k c_k \cdot w_{view,k}$, This forms the analytical basis for weight-attribution waterfalls. Idzorek (2007) converts the confidence percentage c_k into $\Omega_{kk} = [(1-c_k)/c_k]\cdot p_k'(\tau\Sigma)$, here stressed over $\tau \in [0.001, 0.5]$.

Kolm and Ritter (2017) formalise the posterior as the exact solution of a well posed Bayesian inference problem, where τ is interpreted as the ratio of equilibrium to estimation uncertainty. Implementation pathologies (e.g. ill-conditioning for small τ) are enumerated by Drobetz (2001), and are addressed here by dual-form verification and systematic condition-number monitoring. Two extensions are worth mentioning: Bertsimas, Gupta, and Paschalidis (2012) reformulate BL as inverse optimisation, with equilibrium returns as coefficients that justify observed benchmark weights, while Huynh and Ta (2025) replace subjective view specification with GARCH-EVT-Copula–LSTM forecasts. The factor-informed view structure used here strikes a defensible middle ground: it preserves the reproducibility that is absent in subjective views, yet still preserves a transparent factor-to-tilt mapping that is obscured by black-box forecasts.

3.2 Covariance Estimation, Shrinkage and Random Matrix Theory

Estimator quality is critical. The condition number $\kappa(\Sigma)$ determines the stability of the quadratic program and also the economic reasonability of the resulting weights. Ledoit and Wolf (2004) build the linear shrinkage estimator $\hat{\Sigma}_{LW} = \alpha\cdot F + (1-\alpha)\cdot S$ with α selected to minimise the expected Frobenius loss; their single-factor target (Ledoit and Wolf, 2003) models the pricing structure. Ledoit and Wolf (2017) generalise this to analytical nonlinear shrinkage, with transformations depending on eigenvalues and penalising noise eigenvalues more than signal eigenvalues. Their (2022) review unifies both regimes. The random matrix theory offers an alternative spectral paradigm: Bun, Bouchaud and Potters (2017) employ the Marchenko–Pastur law to identify the eigenvalues below $\lambda_+ = \sigma^2(1 + \sqrt{(N/T)})^2$as noise and replace them with their mean under RMT denoising.

The Oracle Approximating Shrinkage (OAS) estimator is obtained by Chen, Wiesel, Eldar and Hero (2010) using only the observed moments. However, this literature characterises the quality of the estimators in terms of statistical loss metrics on the matrix itself, and does not describe how the spectral properties are amplified through the successive inversions embedded in the BL posterior and the quadratic program. These operations scale as inverse eigenvalues, such that a trailing eigenvalue statistically negligible in Frobenius norm can dominate the terminal allocation, a transmission mechanism this study isolates and ties to a critical condition-number threshold.

3.3 Fama–French Factor Models and Cross-Sectional Tests

Motivated by the dividend-discount identity that relates valuation ratios to expected profitability and investment, Fama and French (2015) extend their three-factor model (Fama & French, 1993) with RMW and CMA. This specification guides factor construction, beta estimation and cross-sectional testing here. The Fama and MacBeth (1973) two-pass procedure estimates factor betas in the first pass and prices risk in the second pass. We extend this methodology with rolling 63/126/252-day windows and Shanken (1992) errors-in-variables corrections that inflate second pass standard errors to account for first pass beta noise. The GRS statistic of Gibbons, Ross and Shanken (1989) is an exact finite sample F-test for joint alpha significance. The model is augmented with momentum (UMD; Carhart, 1997) and betting-against-beta (BAB; Frazzini & Pedersen, 2014), exploiting the empirically flatter-than-CAPM

security market line due to leverage constraints. Both are dual-purpose: they are used as attribution regressors and as inputs to the relative-view structure through factor-implied return spreads.

3.4 Extensions to Mean–Variance, Resampling and Higher Moments

Michaud (1989) shows that practical MV optimisation is an error-maximiser. His resampled efficiency solution averages optimal weights over Monte Carlo draws of the input moments, but does not correct the underlying estimation error, and has no coherent Bayesian justification. DeMiguel, Garlappi and Uppal (2009) show that when the estimation error is large compared to the true Sharpe dispersion the 1/N portfolio often performs better than the optimised allocations, motivating its role as the null benchmark. Rockafellar and Uryasev (2000) generalise variance to Conditional Value-at-Risk, a coherent (subadditive, monotonic, positively homogeneous, translation-invariant) risk measure whose convexity allows a tractable linear-programme formulation, implemented here at the 5% level. Jondeau and Rockinger (2006) extend the selection to the MVSK objective. $U(w) = w'\mu - (\delta/2)w'\Sigma w + (\gamma_3/3!)w'M_3(w\otimes w) - (\gamma_4/4!)w'M_4(w\otimes w\otimes w)$that values positive skewness and penalises excess kurtosis through the coskewness and cokurtosis tensors. The resulting non-convex program is solved using SLSQP with fifty Dirichlet seeded restarts to alleviate local optima.

3.5 Risk Parity and Equal Risk Contribution Diversification

Risk parity is appealing in the case of uncertain return estimates, since it does away with expected-return inputs. Maillard, Roncalli, and Teiletche (2010) define the Equal Risk Contribution portfolio via $CRC_i = w_i\cdot(\Sigma w)_i/\sigma_p$, showing it lies between minimum-variance and equal-weight in concentration and maximises entropy of risk contributions; both Spinu (2013) convex reformulation and the Maillard SQP approach are implemented here. Choueifaty and Coignard (2008) propose the diversification ratio $DR = w'\sigma/\sqrt{w'\Sigma w}$ and the Maximum Diversification Portfolio. Meucci, Santangelo, and Deguest (2015) define the effective number of bets via Shannon entropy of minimum-torsion eigenvalue contributions —measuring statistically independent risk sources in factor space rather than raw asset space—enabling the multi-dimensional diversification assessment adopted here. Roncalli (2013) provides the machinery for the marginal, component and percentage risk-contribution underlying the attribution waterfalls, Lorenz curves and Gini calculations.

3.6 Synthesis and Outstanding Limitations

The literature is fragmented, but mature, and five gaps are left open. First, the interaction between the construction of the BL posterior and the choice of covariance estimator has received little systematic attention: canonical implementations (Black & Litterman, 1992; He & Litterman, 1999; Idzorek, 2007) uniformly use the sample covariance, and even Kolm and Ritter (2017) do not propagate shrinkage or RMT alternatives through $\tau\Sigma$, Ω, and Σ_{BL} to terminal weights. Second, factor integration in BL has been ad hoc where factors were either treated as post-hoc attribution regressors (Fama & French, 2015; Carhart, 1997) or as view inputs (Huynh & Ta, 2025), but never both. Third, the higher-moment literature (e.g., Jondeau & Rockinger, 2006; Rockafellar & Uryasev, 2000) has not been incorporated in a BL posterior. Fourth, engineered regime-detection features are built independently of BL. Fifth, and most pervasively, evaluation frameworks rely on in-sample or single-split windows that ignore structural regime shifts the BL literature validating within calibration periods, factor pricing invoking full-sample GRS tests under stationarity assumptions, and even Huynh and Ta (2025) imposing no strict temporal separation. These gaps are the motivation for the integrative framework developed herein.

4. Research Gap

4.1 Covariance Propagation through the Black–Litterman Pipeline not Characterised

The canonical implementations of the Black–Litterman (BL) model (Black & Litterman, 1992; Idzorek, 2007; Kolm & Ritter, 2017) assume a single covariance specification and do not address the sensitivity of the posterior to alternative estimators. This omission is important because Σ enters the posterior in four algebraically nested channels: the equilibrium prior $\Pi = \delta\Sigma\tilde{w}$; the prior-uncertainty scaling $\tau\Sigma$; the Idzorek view-uncertainty calibration $\Omega_{kk} = [(1-c_k)/c_k]\cdot p_k'(\tau\Sigma)p_k$, where each view's precision is a quadratic function of Σ; and the posterior covariance $\Sigma_{BL} = [(\tau\Sigma)^{-1} + P'\Omega^{-1}P]^{-1}$, whose inversion is governed by $\kappa(\Sigma)$. The channels interact multiplicatively, in nested inversions. A perturbation in Σ simultaneously shifts the equilibrium anchor, rescales prior precision, recalibrates the uncertainty of each view, and changes posterior dispersion. Terminal weights are an uncharacterised nonlinear function of estimator choice. This work propagates five estimators (Sample κ=548, Ledoit–Wolf κ=5.9, RMT κ=509, OAS κ=485, nonlinear shrinkage κ=528) through the full pipeline over three risk-aversion regimes with B=1,000 bootstrap resamples, isolating a critical condition-number threshold at $\kappa\approx10$ that divides estimators into structurally distinct allocation regimes.

4.2 Piecewise Factor Integration

Existing work treats factor integration as a one step exercise in universe construction, view generation or ex-post attribution. Therefore, a tilt introduced at one step is routinely nullified by constraints at another. The tripartite Universe–Screen–Diagnostic methodology of this paper embeds factor exposures at admission (factor-mimicking ETFs and synthetic long–short portfolios), gates candidates on rolling Vasicek-shrunk betas $\beta_{V} = (\sigma^2_{cs}\cdot\beta_i + \sigma^2_i\cdot\bar{\beta})/(\sigma^2_{cs} + \sigma^2_i)$, and validates them ex-post through seven-factor regressions (Mkt-RF, SMB, HML, RMW, CMA, UMD, BAB) with Newey–West HAC errors and Shanken EIV corrections, thereby enforcing end-to-end tilt consistency.

4.3 Higher-Moment and Bayesian-Return Preferences

In Disconnected Form Higher-moment optimisation (Jondeau & Rockinger, 2006; Rockafellar & Uryasev, 2000) uses raw sample moments, without Bayesian regularisation, while BL uses posterior means in mean–variance objectives that ignore distributional asymmetry. The gap is filled by this paper in providing μ_{BL} as the first-moment input to $U(w) = w'\mu_{BL} - (\delta/2)w'\Sigma w + (\gamma_3/3!)w'M_3(w\otimes w) - (\gamma_4/4!)w'M_4(w\otimes w\otimes w)$, with $N\times N^2$ coskewness and $N\times N^3$ cokurtosis tensors solved with SLSQP with fifty Dirichlet restarts, the first joint imposition of Bayesian return regularisation and tail-asymmetry aversion in a single cohesive objective.

4.4 Out-of-Sample Validation without Regime Knowledge

The literature reviewed does not impose a strict temporal separation between calibration and evaluation across structurally heterogeneous regimes. The study trains on 2000-2020 (dot-com, GFC, European sovereign crisis, COVID-19) and tests on 2022-2025 (aggressive Fed tightening, March 2023 banking stress, AI-driven rally), structurally avoiding look-ahead bias and overfitting

5. Research Flow and Methodological Mapping

The candidate universe is constructed to meet three jointly constraining admissibility constraints for the Black–Litterman and factor machinery downstream: breadth of uncorrelated risk sources, embedded factor structure, and estimator well-posedness. Ingestion (Table 0) is deliberately over-inclusive across nine asset-class groupings – US sector and broad equities, developed and emerging international equities, nominal and inflation-linked fixed income, credit, commodities, real estate, currency and cryptocurrency – so that the $|\rho| < 0.70$ prune and Marchenko–Pastur screen operate on economically

independent factors rather than near-collinear equity proxies. Embedded exposures (VLUE, MTUM, QUAL, SIZE, OMFL; SPLV/USMV vs SPHB for BAB; HYG, EMB, SCHD for carry) maintain tilt consistency from selection through attribution. Commodities and the dollar enhance covariance conditioning, and cryptocurrencies purposely stress the shrinkage estimators.

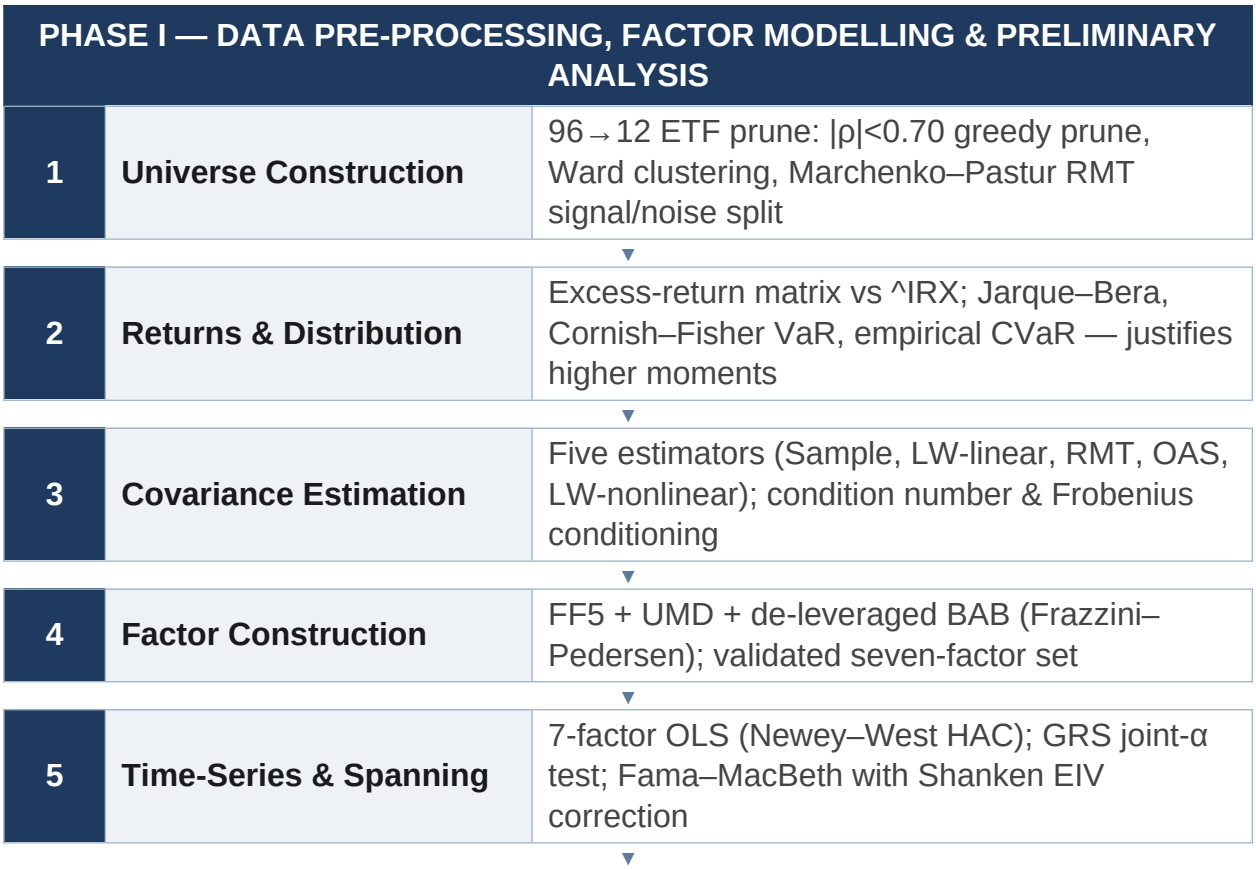
Table 0. Variable List — candidate ETF universe ingested at Stage 1 (63 instruments across nine asset-class groupings), prior to correlation, clustering, and spectral screening.

Ticker	Instrument / Fund Name	Asset Class / Factor
XLK	Technology Select Sector SPDR Fund	US Equities (Sector)
XLV	Health Care Select Sector SPDR Fund	US Equities (Sector)
XLF	Financial Select Sector SPDR Fund	US Equities (Sector)
XLY	Consumer Discretionary Select Sector SPDR Fund	US Equities (Sector)
XLP	Consumer Staples Select Sector SPDR Fund	US Equities (Sector)
XLI	Industrial Select Sector SPDR Fund	US Equities (Sector)
XLE	Energy Select Sector SPDR Fund	US Equities (Sector)
XLU	Utilities Select Sector SPDR Fund	US Equities (Sector)
XLRE	Real Estate Select Sector SPDR Fund	US Equities (Sector)
XLB	Materials Select Sector SPDR Fund	US Equities (Sector)
XLC	Communication Services Select Sector SPDR Fund	US Equities (Sector)
SPY	SPDR S&P 500 ETF Trust	US Equities (Broad)
QQQ	Invesco QQQ Trust	US Equities (Broad)
DIA	SPDR Dow Jones Industrial Average ETF Trust	US Equities (Broad)
IWM	iShares Russell 2000 ETF	US Equities (Broad)
VTI	Vanguard Total Stock Market ETF	US Equities (Broad)
VUG	Vanguard Growth ETF	US Equities (Broad)
VTV	Vanguard Value ETF	US Equities (Broad)
RSP	Invesco S&P 500 Equal Weight ETF	US Equities (Broad)
SCHD	Schwab US Dividend Equity ETF	US Equities (Broad / Equity Carry)
EFA	iShares MSCI EAFE ETF	Intl Developed Equities
VEA	Vanguard FTSE Developed Markets ETF	Intl Developed Equities
IEFA	iShares Core MSCI EAFE ETF	Intl Developed Equities
EZU	iShares MSCI Eurozone ETF	Intl Developed Equities
EWJ	iShares MSCI Japan ETF	Intl Developed Equities
VGK	Vanguard FTSE Europe ETF	Intl Developed Equities
EWU	iShares MSCI United Kingdom ETF	Intl Developed Equities
VWO	Vanguard FTSE Emerging Markets ETF	Emerging Markets
IEMG	iShares Core MSCI Emerging Markets ETF	Emerging Markets
EEM	iShares MSCI Emerging Markets ETF	Emerging Markets
MCHI	iShares MSCI China ETF	Emerging Markets
INDA	iShares MSCI India ETF	Emerging Markets
EWZ	iShares MSCI Brazil ETF	Emerging Markets
EWT	iShares MSCI Taiwan ETF	Emerging Markets
AGG	iShares Core US Aggregate Bond ETF	Fixed Income
BND	Vanguard Total Bond Market ETF	Fixed Income
TLT	iShares 20+ Year Treasury Bond ETF	Fixed Income
IEF	iShares 7-10 Year Treasury Bond ETF	Fixed Income
SHY	iShares 1-3 Year Treasury Bond ETF	Fixed Income
LQD	iShares iBoxx $ Investment Grade Corporate Bond ETF	Fixed Income
HYG	iShares iBoxx $ High Yield Corporate Bond ETF	Fixed Income (Credit Carry)
BNDX	Vanguard Total International Bond ETF	Fixed Income
TIP	iShares TIPS Bond ETF	Fixed Income
GLD	SPDR Gold Shares	Alternatives

Ticker	Instrument / Fund Name	Asset Class / Factor
SLV	iShares Silver Trust	Alternatives
USO	United States Oil Fund	Alternatives (Oil / Commodity Proxy)
DBC	Invesco DB Commodity Index Tracking Fund	Alternatives
VNQ	Vanguard Real Estate Index Fund	Alternatives
VNQI	Vanguard Global ex-U.S. Real Estate ETF	Alternatives
UUP	Invesco DB US Dollar Index Bullish Fund	Alternatives (DXY Proxy)
VLUE	iShares MSCI USA Value Factor ETF	Equity Factors (Value)
MTUM	iShares MSCI USA Momentum Factor ETF	Equity Factors (Momentum)
QUAL	iShares MSCI USA Quality Factor ETF	Equity Factors (Quality)
SIZE	iShares MSCI USA Size Factor ETF	Equity Factors (Size)
SPLV	Invesco S&P 500 Low Volatility ETF	Equity Factors (Low Vol / Long BAB)
USMV	iShares MSCI USA Min Vol Factor ETF	Equity Factors (Min Vol / Long BAB)
SPHB	Invesco S&P 500 High Beta ETF	Equity Factors (High Beta / Short BAB)
OMFL	Invesco Russell 1000 Dynamic Multifactor ETF	Equity Factors (Multi-Factor)
EMB	iShares J.P. Morgan USD Emerging Markets Bond ETF	Fixed Income (EM Carry)
BTC-USD	Bitcoin USD	Cryptocurrency
ETH-USD	Ethereum USD	Cryptocurrency
CL=F	Crude Oil Futures	Commodities
DX-Y.NYB	US Dollar Index (DXY)	Currencies

Below, in Figure 0, we provide the full analytical architecture linking the data infrastructure of Section 4 to the empirical findings of Section 7. It diagrams the seventeen-stage pipeline across the three macro-phases before the narrative that follows justifies each transition on econometric grounds.

Figure 0. Seventeen-Stage Factor-Augmented Black–Litterman Research Pipeline



6	**Regime & Structural Breaks**	Rolling betas; Hamilton 2-state Markov switching; Bai–Perron & PELT changepoint detection
▼		
7	**Factor Introduction**	Tripartite Universe / Screen / Diagnostic methods; Vasicek Bayesian beta shrinkage
▼		
8	**Baseline Backtesting**	Equal-weight & inverse-volatility nulls; 19-metric suite; Deflated Sharpe Ratio
▼		
PHASE II — BLACK–LITTERMAN POSTERIOR, MULTI-OBJECTIVE OPTIMISATION & SHRINKAGE		
9	**BL Prior (Reverse Opt.)**	Equilibrium returns $\Pi = \delta\Sigma\tilde{w}$ from cap-weight proxy across three risk-aversion regimes
▼		
10	**BL Posterior (Bayesian Update)**	Pick matrix P, view vector Q, Idzorek Ω calibration; dual precision-weighted / Kalman-gain forms
▼		
11	**Multi-Objective Optimisation**	Constrained MVO, Dinkelbach Max-SR, Rockafellar–Uryasev Min-CVaR, Athayde–Flores MVSK (CVXPY cascade)
▼		
12	**Covariance Sensitivity**	Re-optimise under 5 Σ estimators; B=1,000 bootstrap; resolves stability paradox
▼		
PHASE III — RISK DECOMPOSITION, DIVERSIFICATION & ATTRIBUTION		
13	**Master Comparison**	12 portfolio variants × 19 metrics; first- & second-order stochastic dominance adjudication
▼		
14	**Euler Risk Decomposition**	MRC / CRC / PRC; diversification ratio, ENB; Spinu convex ERC reformulation
▼		
15	**Out-of-Sample Backtesting**	Walk-forward application of optimised weights; wealth & drawdown trajectories
▼		
16	**SR Significance**	Jobson–Korkie–Memmel z-test; autocorrelation-adjusted bootstrap confidence intervals
▼		
17	**Factor Attribution & FLAM**	Nested multi-factor OLS; Treynor–Black appraisal ratio; α vs factor decomposition

Figure 0. Architectural flowchart of the research workflow. Each numbered box is a deterministic analytical stage whose output is the sufficient statistic for its successor; downward arrows denote econometric dependencies. Phase I conditions raw data into a well-posed covariance and a validated seven-factor return structure; Phase II fuses these through Bayesian updating and competing optimisation objectives; Phase III decomposes risk and validates performance out-of-sample.

In this section we formalise the end-to-end flow from raw market observations to the empirical findings of Section 7, relating each analytical operation to its econometric justification and to the objectives of Section 2. The workflow is a 17-stage pipeline across three sequential macro-phases; the ordering is dictated by analytical, not expository convenience; each stage produces the sufficient statistic that its successor needs, so estimation error is contained at its origin and propagated explicitly rather than compounded silently through the nested matrix inversions that dominate downstream optimisation. The pipeline is inherently auditable since each Section 6 concept is associated with a numbered stage, and each Section 7 exhibit inherits unambiguous provenance, which is a contribution in a field where reported performance is notoriously sensitive to undocumented latitude in pre-processing.

Phase I — Conditioning Raw Data into Well-Formed Inputs

Stage 1 reduces a ninety-six-ETF universe to twelve constituents by governed means: a missing-observation screen, a greedy Pearson prune preserving pairs below |0.70|, and Ward-linkage clustering on the ultrametric distance $d = \sqrt{(2(1-\rho))}$, with the Marchenko–Pastur upper spectral edge setting latent dimensionality non-parametrically and Shannon-entropy breadth confirming seven surviving eigenvalues. The ex-ante guarantee of well-conditioning pre-empts the spectral ill-posedness otherwise propagated through $\Sigma-1$. Stage 2 yields excess returns over the matched three-month T-bill; the downstream Cornish–Fisher and empirical Expected-Shortfall mechanisms are justified by universal Jarque–Bera rejection of Gaussianity. In Stage 3, the covariance estimator is promoted from a fixed assumption to a controlled experimental variable. Five estimators (Sample, Ledoit–Wolf linear, RMT-denoised, OAS, nonlinear shrinkage) are benchmarked on condition number, Frobenius distance, and spectral structure.

Phase II — The Inference Core

Stage 9 eliminates the error maximisation pathology of unconstrained MVO at source, reverse optimising over three risk aversion regimes to recover equilibrium returns $\Pi = \delta\Sigma\tilde{w}$. Stage 10: Bayesian update. Factor-informed views are represented by a pick matrix with Idzorek-mapped uncertainty. Posterior is recovered in dual precision weighted and Kalman gain forms subject to numerical identity verification. Stage 11 runs the posterior through mean-variance, Dinkelbach maximum-Sharpe, Rockafellar–Uryasev minimum-CVaR and Athayde–Flores MVSK programs via a CVXPY cascade. Stage 12 finishes the covariance experiment with B = 1,000 bootstrap resampling, solving the stability paradox: allocation robustness is determined by conditioning, not point accuracy.

Phase III — Translating Allocations into Defensible Inference

Stage 13 applies first- and second-order stochastic dominance rankings to twelve portfolio variants across a nineteen-metric scorecard – utility-free adjudication that survives the non-normality of Stage 2. Stage 14 decomposes Euler into marginal, component and percentage risk contributions. Diversification ratios and effective bets are computed and equal-risk-contribution allocations are derived using Spinu's convex reformulation, separating weight-based and risk-based diversification. Stages 15–17 conduct walk-forward out-of-sample backtests, arbitrate Sharpe differentials via the Jobson–Korkie–Memmel test with autocorrelation-adjusted bootstrap intervals, and conclude with multi-factor attribution and Fundamental-Law diagnostics that decompose realised performance into idiosyncratic alpha and residual factor components—to answer the ultimate question of whether outperformance is a function of skill or mechanical factor replication.

The architecture is thus a monotonic cascade of information refinement in which each terminal exhibit of Section 7 is derivable from its econometric antecedent.

6. Research Methodology

We propose a Factor-Augmented Black–Litterman (FABL) portfolio construction framework based on sixty-seven methodological tools, covering fifteen themes. All of them are part of the inferential integrity, but there are five instruments that are the load-bearing pillars, without them the architecture collapses. Each was chosen according to the criterion of causal centrality in the seventeen stage pipeline: it either changes problem dimensionality, stabilises an otherwise intractable estimation or enables a claimed novel contribution.

6.1 Linear Shrinkage Covariance Estimator: Ledoit–Wolf (2004)

Covariance fidelity is the most important decision an estimator faces. It appears in the BL posterior in four places: equilibrium returns ($\Pi = \delta\Sigma\tilde{w}$), prior uncertainty ($\tau\Sigma$), view uncertainty ($\Omega$), and posterior covariance. The Ledoit–Wolf estimator is a convex combination of the sample covariance S and a structured target F:

$\Sigma_LW = (1 - \delta)S + \delta F$, $\delta^* = \text{argmin } E[\|\Sigma_LW - \Sigma_true\|^2_F]$ where δ^* is the analytically optimal shrinkage intensity that minimises the expected Frobenius loss. The bias-variance story is very clear, F is biased but perfectly conditioned, S is unbiased but has high variance in its smallest eigenvalues. In this study Ledoit–Wolf attains $\kappa = 5.93$ compared to $\kappa = 548.31$ for the sample estimator. This is a two orders of magnitude compression crossing the critical $\kappa \simeq 10$ threshold below which the MVO operator activates all the twelve assets instead of six. Bootstrap analysis (B = 1,000) yields a Sharpe SD of 0.0282 vs. 0.0910, a 3.2-fold reliability gain that resolves the stability paradox: the most structurally dissimilar portfolio is the most robust. Advantages are closed-form intensity, simplicity of computation and guaranteed positive semi-definiteness. The main drawback is the uniform contraction of eigenvalues toward a common target, removing spacing of the spectrum ($N_eff = 9.51$ vs. 3.35 for Sample) — the price of conditioning at the cost of factor fidelity.

6.2 Black–Litterman Posterior (Precision-Weighted and Kalman-Gain Forms)

The BL posterior transforms unstable historical means, the Achilles heel of MVO (Michaud, 1989), into an economically justified expected-return vector. The posterior mean has two algebraically equivalent expressions. The precision-weighted form is $\mu_BL = [(\tau\Sigma)^{-1} + P'\Omega^{-1}P]^{-1}\,[(\tau\Sigma)^{-1}\Pi + P'\Omega^{-1}Q]$ and the Kalman-gain mixed-estimator form is $\mu_BL = \Pi + \tau\Sigma P'[P\tau\Sigma P' + \Omega]^{-1}(Q - P\Pi)$ where τ scales prior uncertainty, $\Pi = \delta\Sigma\tilde{w}$ is the CAPM-reverse-optimised equilibrium prior, Q is the $K \times 1$ view vector, P is the $K \times N$ pick matrix, and Ω is calibrated via Idzorek's mapping $\Omega_kk = [(1 - c_k)/c_k]\cdot p_k'(\tau\Sigma)p_k$.

The relative precision positions the posterior on a spectrum between the view point ($\Omega \to 0$) and the equilibrium prior ($\Omega \to \infty$). Both forms are implemented and verified to numerical identity to machine precision. After three factor-informed views (K = 3), the posterior–equilibrium correlation $\rho(\Pi, \mu_BL) = 0.837$, absorbing a mean 2.835% view adjustment while remaining prior-anchored. Mahalanobis diagnostics show joint-view interference; τ-grid sensitivity over $\tau \in [0.001, 0.5]$ shows posterior stability. Bayesian regularisation damps corner solutions of naive estimation. Limitations: the posterior aggregates errors from all four inputs, the diagonal Ω disregards view correlation, and subjective confidence calibration, though internally consistent under Idzorek, is still necessary.

6.3 Higher-Moment MVSK Utility (Athayde–Flores Co-Moment Tensors)

The universal Jarque–Bera rejection (JB from 1,427 to 1.44×10^8) formally invalidates the Gaussian mean–variance objective. The Athayde–Flores utility maximises the first four moments jointly:

$$U(w) = \lambda_1 w'\mu_{\text{BL}} - \lambda_2 w'\Sigma w + \lambda_3 w' M_3 (w \otimes w) - \lambda_4 w' M_4 (w \otimes w \otimes w)$$

where M_3 is the $N \times N^2$ coskewness tensor and M_4 the $N \times N^3$ cokurtosis tensor. The non-convex program is solved using SLSQP with fifty Dirichlet-seeded random restarts to escape local optima. We

are the first to use BL-posterior means μ_BL as the first-moment input to MVSK. The resulting higher-moment portfolio has the highest expected return (10.5%) among BL-augmented variants and second-order stochastically dominates the minimum-CVaR allocation - a preference-free result that implies that every risk-averse investor, regardless of the utility functional form, would prefer it. Merits: MVSK captures asymmetry and tail fatness invisible to mean-variance; Dirichlet restarts provide reasonable landscape coverage. There are three limitations: high tensor dimensionality (N×N^3=12×1728 for M_4), fast noise growth in tensor order, and no guarantee of global optimality and restarts, as well as outlier sensitivity in tensor estimation.

6.4 Conditional Value-at-Risk (Rockafellar–Uryasev Optimisation)

CVaR (Expected Shortfall) is defined as

$$\mathrm{CVaR}_\alpha = \mathbb{E}\left[X \mid X \le \mathrm{VaR}_\alpha\right] = \frac{1}{\alpha}\int_0^\alpha \mathrm{VaR}_u \, du$$

CVaR, in contrast, is coherent (subadditive, monotonic, positive homogeneous, translation invariant) and therefore aggregation at the portfolio level is well-defined. Rockafellar and Uryasev (2000) show that minimisation is a linear program with auxiliary variable α:

$$\min_{w,\alpha} \ \alpha + \frac{1}{(1-\beta)J}\sum_{j=1}^{J}\left[-w' r_j - \alpha\right]^+$$

where J is the number of return scenarios. Introducing α as a decision variable, the non-smooth objective is converted into a convex tractable program with global optimum. The minimum-CVaR portfolio at 5% obtains the lowest tail risk, CVaR = 8.2%, at the cost of the lowest return (3.4%) and Sharpe (0.2), quantifying the irreducible cross-objective trade-off, and serving as the SSD comparator against the MVSK portfolio. Merits: convex, direct sensitivity of tails, coherency. Limitations: with only 5% × 1,509 = 75 tail observations, the CVaR estimation is considerably more prone to sampling error than variance-based measures, and the scenario-based LP is sample-dependent.

6.5 Euler Risk Decomposition

The exact additive decomposition of portfolio volatility (positively homogeneous of degree one in weights) is obtained by the Euler homogeneity.

$$\sigma_p = \sum_i w_i \frac{\partial \sigma_p}{\partial w_i} = \sum_i \mathrm{CRC}_i, \quad \mathrm{MRC}_i = \frac{(\Sigma w)_i}{\sigma_p}, \quad \mathrm{PRC}_i = \frac{\mathrm{CRC}_i}{\sigma_p}$$

where MRC = marginal risk contribution, CRC = $w_i \cdot \mathrm{MRC}_i$ = component contribution, and PRC = percentage contribution. The identity Σ i CRC i = σ p holds numerically for all twelve variants. The ERC portfolio is calculated via the convex reformulation of Spinu (2013) min_y ½y'Σy - κΣ_i ln y_i, w = y/1'y, which ensures a unique risk-parity fixed point. The analysis shows systematic divergence between diversification by weight and by risk. Risk-parity drawdowns are -6.8% to -16.1% versus -50.77% for equal-weight. ERC is the only approach achieving machine-precision contribution equality (CRC CoV = 0.000) with Ledoit–Wolf and BL-posterior covariances, demonstrating that shrinkage is a numerical prerequisite to achieve exact risk parity. Advantages: exact additivity, economic interpretability and a risk-parity allocation based on principle. Limitations: risk attribution depends on sigma, volatility; tail-, skewness-, and regime-conditional decompositions need CVaR or higher moment extensions not pursued here.

6.6 Synthesis

The five instruments form the inferential backbone of the FABL framework. Ledoit–Wolf conditions the covariance feeding each downstream computation; the BL posterior regularises noisy means to an

equilibrium anchor; MVSK and CVaR extend selection beyond the formally rejected Gaussian; Euler decomposition audits the resulting risk topology. Each addresses a specific failure mode—ill-conditioning, mean instability, distributional misspecification and risk-attribution opacity—to form a coherent, self-correcting pipeline in which each modelling choice is accountable to a preceding diagnostic. The only commonality is that they all use the covariance matrix, either as an input (Ledoit–Wolf, Euler), as a pass-through (BL), or as a way to generate scenarios (CVaR, MVSK); error in estimating second moments pervades the architecture. This dependence is not hidden but explicitly quantified through the multi-estimator propagation analysis that constitutes the first contribution of this study.

7. Empirical Findings and Analytical Results

7.1 Factor-Aware Asset Universe Construction

A 5% missing-observation screen over 2020-01-01 to 2025-12-31 drops two illiquid ETFs (GEM, FM) from 96 candidates, yielding T = 2,189 × N = 94 (Figure 1). The ratio q=N/T ≈ 0.043 is in the asymptotic regime where sample second moments are admissible.



Figure 1. Missing-data profile across the 96-candidate universe; the red dashed line marks the 5% drop threshold that excludes GEM and FM.

iterative greedy prune on RMT-denoised correlation matrix (|ρ| > 0.70 rejection, mean |ρ| < 0.45) with parallel ward-linkage validation by:

$$d_{ij} = \sqrt{2\,(1 - \rho_{ij})} \tag{1}$$

cuts 94 assets to 12 in five classes, reducing mean |ρ| from ≈ 0.45 to 0.2753 (Figure 2). Ward's minimum-variance clustering on the full panel gives cophenetic correlation 0.72-0.78 and ≥ seven distinct clusters (Fig. 3) - consistent with the RMT signal eigenvalues below.

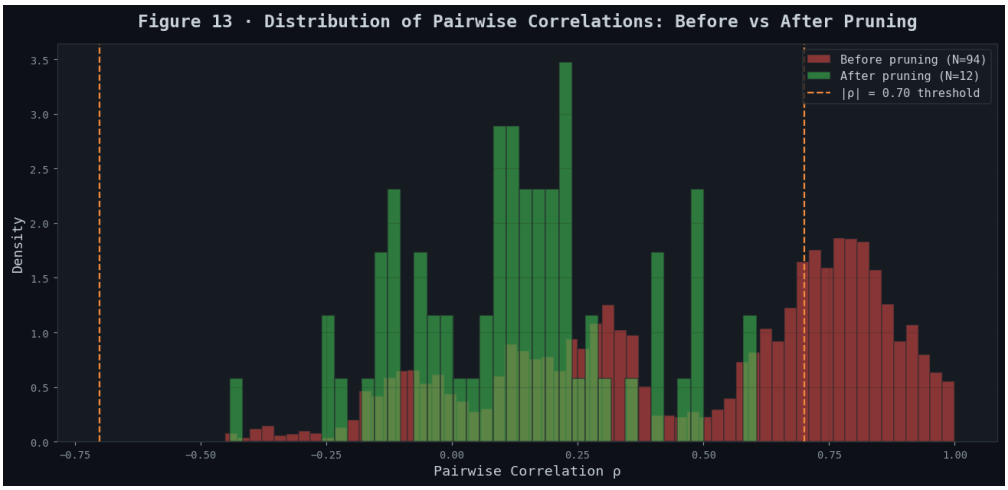


Figure 2. Pairwise-correlation distribution before (N = 94, red) and after (N = 12, green) pruning; dashed verticals mark |ρ| = 0.70.

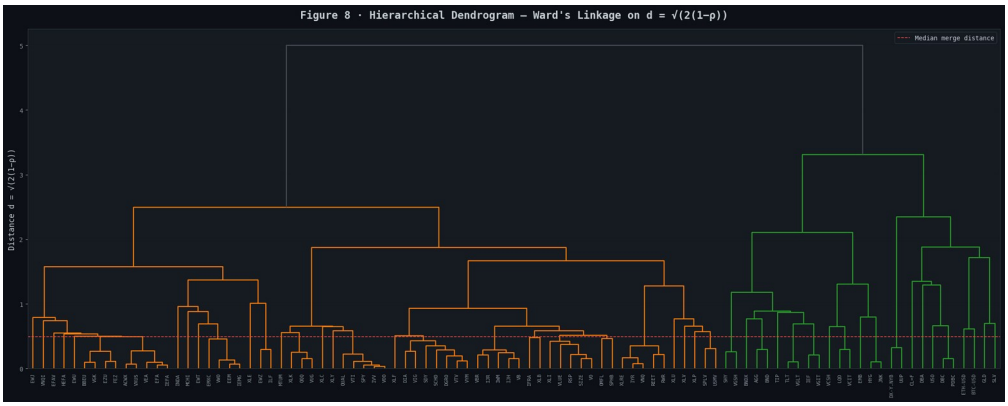


Figure 3. Ward-linkage dendrogram on d = √(2(1−ρ)) for all 94 candidates; the red dashed line marks the median merge distance.

The Marchenko–Pastur bulk edges are:

$$\lambda_{\pm} = (1 \pm \sqrt{q})^2 \tag{2}$$

Seven eigenvalues are above λ+ = 1.4574 (equity-market, duration, commodity, EM, crypto, credit-spread, volatility), the other 87 populate the MP bulk and are eigen-clipped to their mean (Figure 4, Table 1).

Table 1. Marchenko–Pastur RMT Eigenvalue Classification

Parameter	Value	Interpretation
Concentration ratio q = T/N	23.29	Well within asymptotic validity
MP lower bound λ^-	0.6285	Lower edge of noise bulk
MP upper bound λ^+	1.4574	Upper edge of noise bulk
Signal eigenvalues ($\lambda > \lambda^+$)	7	True latent risk factors
Noise eigenvalues ($\lambda \le \lambda^+$)	87	Replaced by their mean
Largest eigenvalue λ_1	≈ 55.2	Market / systematic factor

Note: MP spectral-bound parameters and the count of signal versus noise eigenvalues used to eigen-clip the sample correlation matrix.

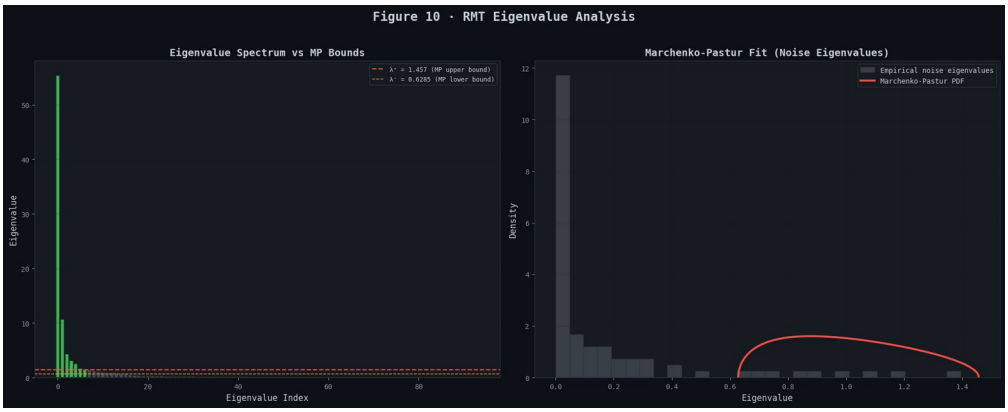


Figure 4. Marchenko–Pastur eigenvalue analysis. Left: signal (green) vs noise (grey) eigenvalues. Right: empirical noise histogram against the theoretical MP density.

Shannon effective dimensionality corroborates:

$$N_{\text{eff}} = \exp\left(-\sum_i p_i \ln p_i\right), \quad p_i = \frac{\lambda_i}{\sum_j \lambda_j} \tag{3}$$

yielding N_eff = 6.53 vs. N = 94 (Table 2), a 93% reduction, with PC1 alone explaining > 55% of variance (Figure 5).

Table 2. Shannon Entropy Dimensionality Analysis

Metric	Value	Interpretation
Shannon entropy H	1.8772	Low entropy → concentrated structure
Effective dimensionality N_eff	6.53	~6.5 independent risk sources
Total assets N	94	Nominal dimensionality
Dimensionality reduction ratio	6.9%	93% of dimensions redundant

Note: Information-theoretic dimensionality metrics quantifying redundancy in the 94-asset covariance spectrum.

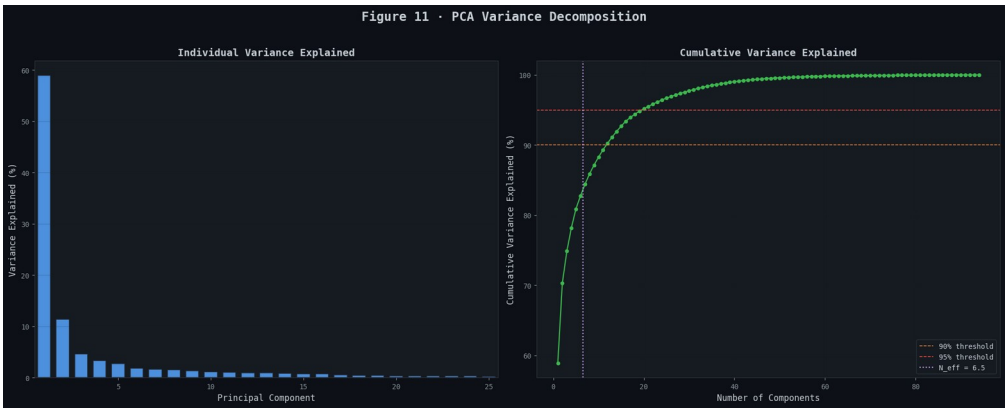


Figure 5. PCA variance decomposition — per-PC and cumulative variance with 80%/95% markers; annotated N_eff = 6.5.

Three independent instruments – Ward clustering, MP spectral analysis, Shannon entropy – converge on the same seven-dimensional risk manifold, providing the 12-asset portfolio an order-of-magnitude improvement in downstream covariance-inversion conditioning at negligible cost in investable variation.

SPY CAPM regressions (Newey–West HAC, lag 12) yield betas ranging from −0.101 (TLT) to 1.405 (ETH-USD). Only GLD and ETH-USD exhibit $|t(\alpha)| > 2.0$ (Table 3). Durbin–Watson is close to 2.0 but Breusch–Pagan rejects homoskedasticity for 9 of 12 assets (Figure 6). The 12×12 factor-beta lattice has VIF < 5 at all points (Figures 7–8).

Table 3. CAPM Regression — Pruned Universe (HAC SEs, lag = 12)

Ticker	α (ann.)	t(α)	β_Mkt	t(β)	R^2	BP p
CL=F	−0.447	−0.88	0.896	2.39	0.017	0.040
GLD	0.115	2.72	0.105	3.48	0.018	0.041
INDA	−0.008	−0.19	0.740	7.48	0.449	0.000
MCHI	−0.042	−0.58	0.705	19.34	0.222	0.058
QQQ	0.020	0.80	1.143	32.34	0.877	0.645
TLT	−0.010	−0.23	−0.101	−1.87	0.015	0.000
USO	−0.048	−0.37	0.589	4.44	0.085	0.000
XLE	0.005	0.06	1.003	12.45	0.359	0.006
XLU	−0.003	−0.06	0.682	8.67	0.404	0.002
ETH-USD	0.440	2.16	1.405	10.45	0.127	0.000

Note: Per-asset CAPM intercept, market beta, fit and Breusch–Pagan heteroskedasticity p-values for the 12 pruned assets under HAC inference.

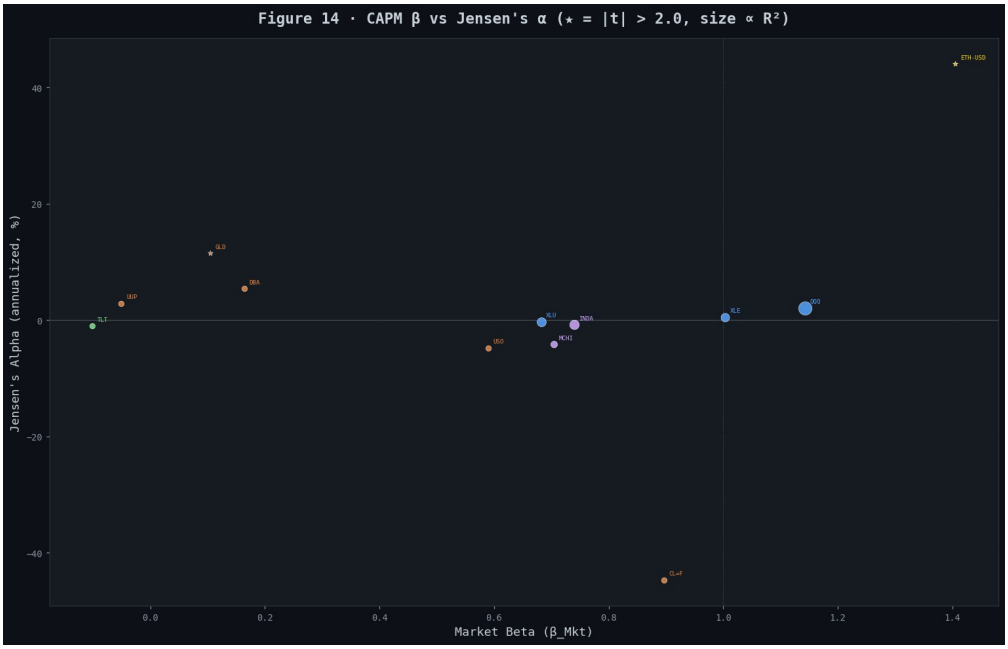


Figure 6. CAPM β versus Jensen α scatter; bubble size ∝ R^2, star markers denote |t| > 2.0.



Figure 7. Bivariate factor-beta heatmap (12 factors × 12 assets, HAC SEs) with significance stars.

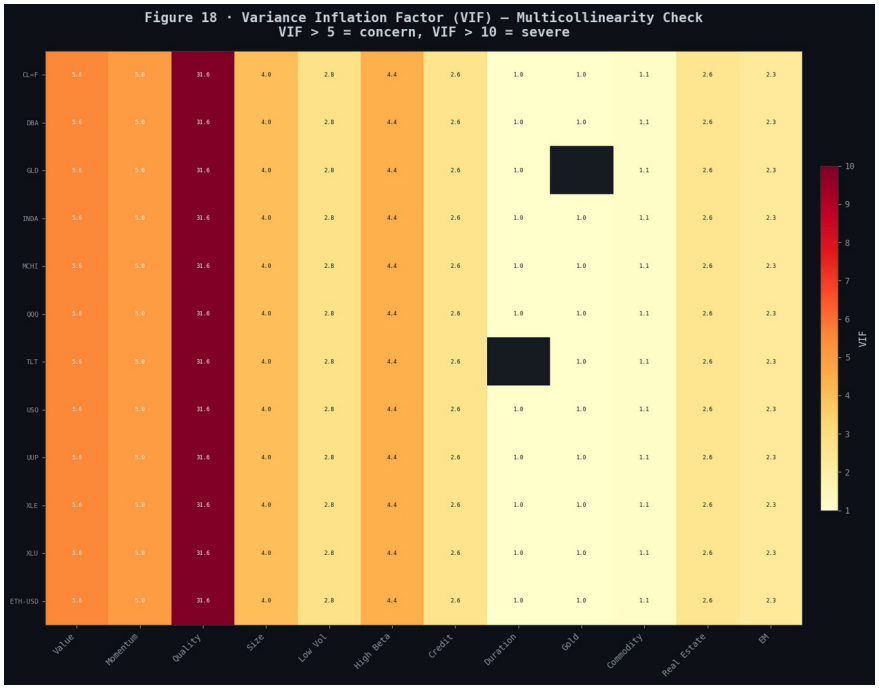


Figure 8. VIF lattice across the 12×12 asset–factor matrix; all cells fall below the VIF = 5 concern threshold.

7.2 Factor Time-Series Construction and Econometric Analysis

The Jarque–Bera statistic:

$$\mathrm{JB} = \frac{T}{6}\left[S^2 + \frac{1}{4}(K-3)^2\right] \sim \chi^2_2 \text{ under } H_0 \tag{4}$$

rejects Gaussianity for all 12 assets (JB ranging from 1,427 to 1.44 × 10^8), all with positive excess kurtosis (Table 4). Gaussian VaR is uninformative for heavy tails (CL=F: −234.8%); empirical CVaR shows real severity (ETH-USD: −150.9%) The §7.8 CVaR and Cornish–Fisher extensions are motivated by universal non-normality (Figures 9–10).

Table 4. Jarque–Bera and Higher-Moment Risk — Pruned Universe (annualised)

Asset	Mean %	Vol %	Sharpe	Skew	Ex.Kurt	JB	CVaR 5%
GLD	9.98	13.56	0.74	−0.35	5.65	2,941	−33.4
QQQ	12.07	21.01	0.57	−0.11	11.50	1.2e4	−52.1
ETH-USD	56.98	67.87	0.84	−0.20	8.34	6,329	−150.9
TLT	−4.81	14.19	−0.34	0.22	7.74	5,459	−33.1
CL=F	−37.36	120.06	−0.31	−32.30	1258.9	1.4e8	−154.8
XLE	8.98	28.81	0.31	−0.55	18.48	3.1e4	−68.7

Note: Annualised moments (mean, vol, Sharpe, skewness, excess kurtosis), Jarque–Bera statistic, and 5% empirical CVaR for representative pruned assets.



Figure 9. Excess-return kernel densities with best-fit Gaussian overlays (dashed red) for each of the 12 assets.

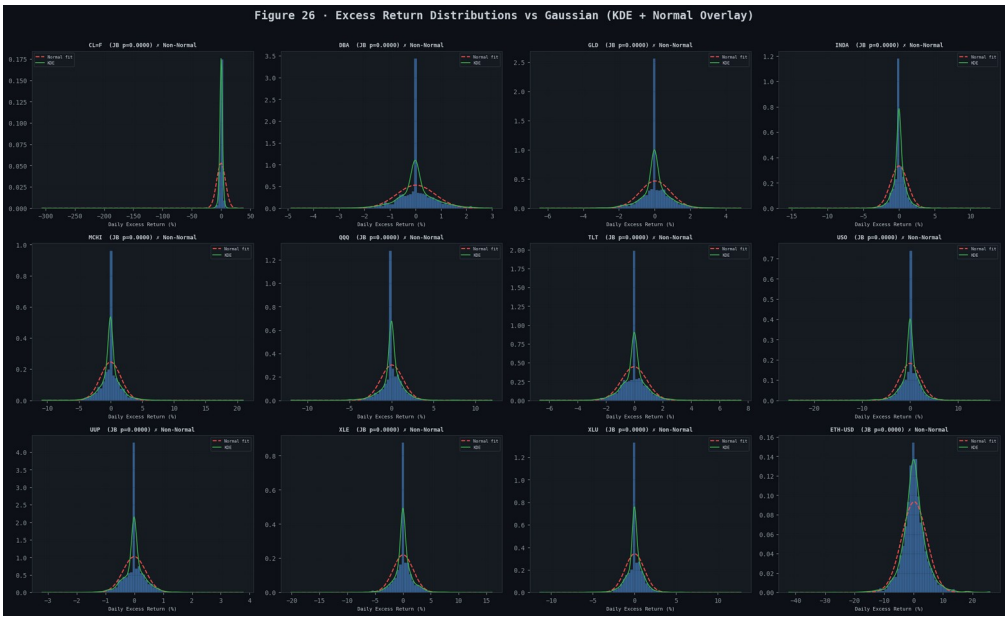


Figure 10. Quantile–quantile plots of excess returns versus the normal distribution; tail departures confirm heavy-tailedness.

Five Fama–French factors (Table 5, max ρ(SMB,RMW)=0.55, Figure 11) plus two proprietary factors: BAB (beta-neutralised, +15.77% p.a., Sharpe 0.133 - EIV fragility at N=12), UMD (12-1 sector momentum, +11.29% p.a., Sharpe 0.516).

Table 5. Fama–French Annualised Descriptive Statistics (%, daily × 252)

Factor	Mean %	Std %	Min %	Max %
Mkt-RF	−1.338	197.82	−1154.16	1438.92
SMB	7.012	263.66	−1267.56	1695.96
HML	2.633	158.16	−561.96	1071.00
RMW	1.531	142.45	−735.84	622.44
CMA	2.835	2.375	0.000	5.040

Note: Annualised mean, standard deviation, and extreme daily observations for each Fama–French factor across the 2020–2025 sample.

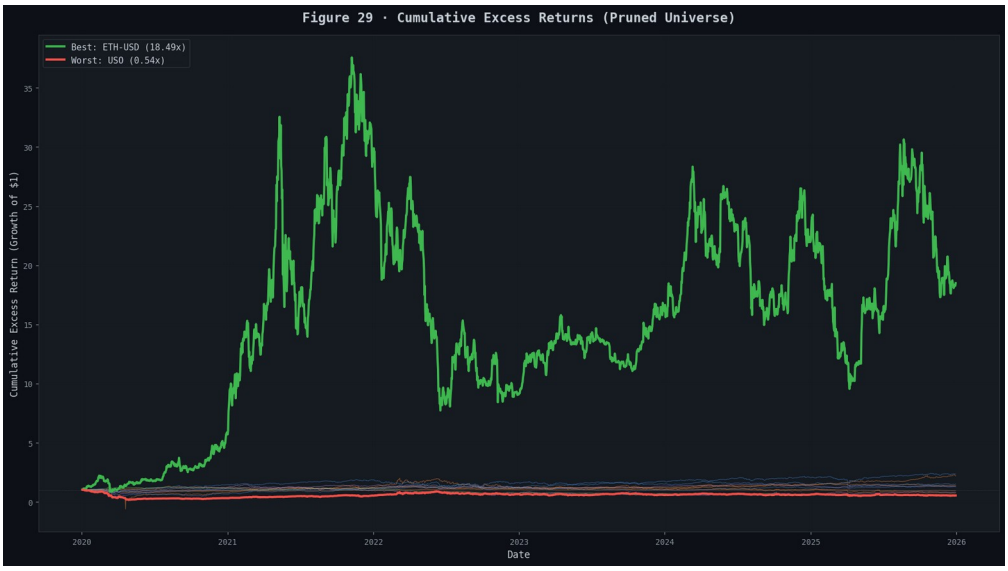


Figure 11. Fama–French five-factor cumulative performance (log-scale) and drawdown profiles, 2020–2025 (Mkt-RF max DD −41.68%).

The seven-factor OLS:

$$r_{i,t}^{\text{ex}} = \alpha_i + \sum_{k=1}^{K} \beta_{i,k} F_{k,t} + \varepsilon_{i,t} \tag{5}$$

with K = 7, HAC lags ⌊T^{1/3}⌋ = 12, gives mean α = 8.21% (16.7% sig. at 5%, dispersion 40.31%; Table 6). The Gibbons–Ross–Shanken (GRS) joint test:

$$\text{GRS} = \frac{T-N-K}{N} \left(1 + \hat{\mu}_F' \hat{\Omega}_F^{-1} \hat{\mu}_F\right)^{-1} \hat{\alpha}' \hat{\Sigma}_\varepsilon^{-1} \hat{\alpha} \sim F(N, T-N-K) \tag{6}$$

gives F = 1.322 (p = 0.199) — cannot reject H_0: α = 0, so the model spans the 12-asset MV frontier (Figure 12). Fama–MacBeth with Shanken adjustment:

$$\text{SE}_{\text{Shanken}} = \text{SE}_{\text{OLS}} \times \sqrt{1 + \kappa^2} \tag{7}$$

inflates SEs by 1.556×, making all premiums insignificant (HML +12.41%, t = 0.597) – a N = 12 power limit, not a failure to reject factors. Ljung-Box does not reject white noise for SMB, HML, CMA, supporting HAC (Table 6).

Table 6. Factor-Model Adequacy — Consolidated Spanning Diagnostics

Diagnostic	Statistic	Value	Inference
BAB (Frazzini–Pedersen)	Sharpe / Mean	0.133 / +15.77%	Low-beta anomaly present
UMD (12−1 momentum)	Sharpe / Mean	0.516 / +11.29%	Momentum profitable
7-factor OLS	Mean α / % sig.	8.21% / 16.7%	Model adequate on avg.
GRS joint test	F / p	1.322 / 0.199	Fail to reject α = 0
Fama–MacBeth (Shanken)	√(1+κ²)	1.556×	All λ insignificant (N power)

Note: Consolidated summary statistics from proprietary-factor performance, OLS pricing regressions, and GRS/Fama–MacBeth spanning tests.



Figure 12. GRS spanning test: individual α t-statistics against the |t| = 1.96 band and the joint F, p-value, fail-to-reject conclusion.

7.3 Covariance Estimation and Conditioning

Downstream sensitivity to second-moment risk is probed by five positive-definite estimators (Table 7). Ledoit–Wolf linear shrinkage:

$$\hat{\Sigma}_{\mathrm{LW}} = (1-\delta^*)S + \delta^*\mu I, \qquad \delta^* = 0.6035 \tag{8}$$

The disciplined nulls are Inverse-Volatility (IV) and 1/N. Equal weight Sharpe 0.1314 (Lo (2002) SE = 0.0372, skew -8.75) IV Sharpe 0.1594 at 8.04% vol. statistically (§7.11) and BL and higher-moment portfolios must beat these.

Table 7. Covariance-Estimator Comparison (annualised, 12×12)

Estimator	δ	κ(Σ)	λ_max	λ_min	N_eff
Sample	0.000	548.31	1.4797	0.00270	3.35
Ledoit–Wolf	0.6035	5.93	0.7042	0.1188	9.51
RMT-denoised	—	509.04	1.4723	0.00289	3.40
OAS	0.0018	484.62	1.4766	0.00305	3.36
Non-linear	indiv.	528.23	1.4468	0.00274	3.36

Note: Shrinkage intensity, condition number, extremal eigenvalues, and effective dimensionality for the five candidate covariance estimators.

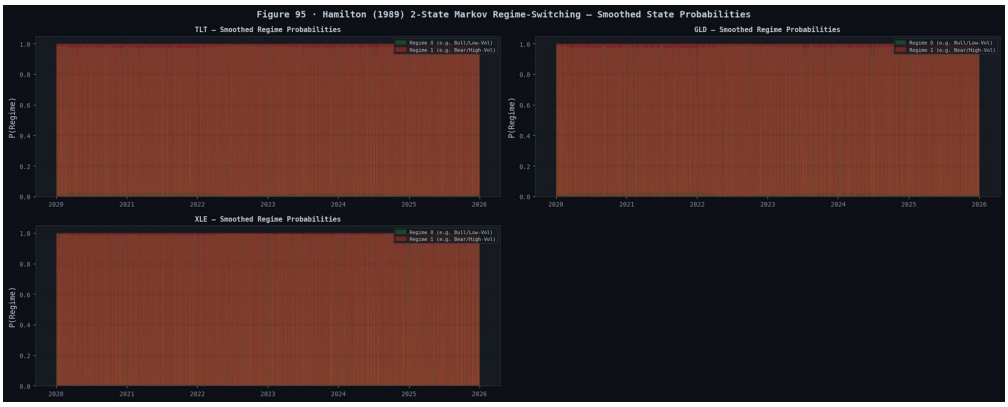


Figure 13. Eigenvalue spectra (log-scale, left) and condition-number/N_eff bar chart (right) across all five estimators.

7.4 Pre-Optimisation Systematic Backtesting

The disciplined nulls are Inverse-Volatility (IV) and 1/N. Equal weight Sharpe 0.1314 (Lo (2002) SE = 0.0372, skew -8.75) IV Sharpe 0.1594 at 8.04% vol. statistically (§7.11) and BL and higher-moment portfolios must beat these.

7.5 Rolling-Beta and Regime-Conditional Factor Analysis

Multi-window rolling OLS (63/126/252-day) captures COVID beta dislocations and 2022 rate-sensitivity drift. Market-neutrality regressions $F_t = \alpha_F + \beta_F(Mkt-RF)_t + \varepsilon_t$ (Table 8) - near-perfect neutrality for RMW ($\beta = 0.040$) and UMD ($\beta = -0.007$); significant negative loading for BAB ($\beta = -0.754$, $t = -2.45$); CMA the only significant α (+2.84% p.a., $t = 15.54$). BAB (max DD −98.97%, Calmar 0.159, Ulcer 88.02, Pain 85.42) is fragile. This is an amplification of de-leveraging and justifies the §7.8 size cap.

Table 8. Factor–Market Neutrality and Drawdown Diagnostics

Factor	α % p.a.	β_Mkt	t(β)	Max DD %	Calmar
RMW	+1.585	0.040	+0.38	—	—
CMA	+2.835	−0.000	−1.41	—	—
BAB	+16.562	−0.754	−2.45	−98.97	0.159
UMD	+11.269	−0.007	−0.13	—	—

Note: Market-neutrality regression intercepts and betas plus drawdown severity metrics for the four non-market factors.

A two-state Hamilton (1989) Markov-switching model cleanly separates bull and bear regimes; GLD is the most stable (AIC/BIC: −18,060/−18,003; Figure 14). PELT over-segments vs BinSeg under BIC penalties (Figure 15). Only concordant breaks are kept. BAB and UMD positive in stress regimes, SMB pro-cyclical in regime-conditional Sharpe. UMD ranks first on composite dashboard.

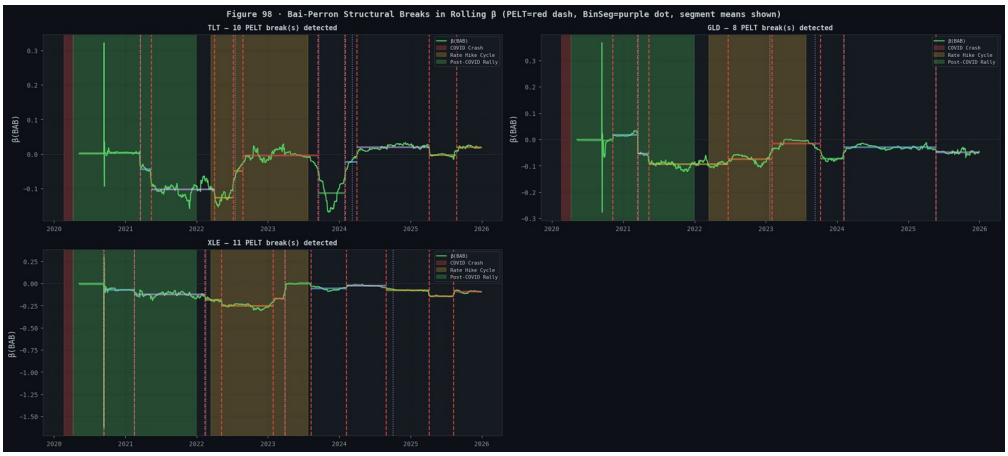


Figure 14. Hamilton (1989) two-state smoothed regime probabilities for GLD, TLT, and a representative equity ETF.



Figure 15. PELT versus BinSeg structural breakpoints on rolling-beta series; only concordant breakpoints retained.

7.6 Systematic Factor Introduction into the Investment Process

Dual-track factors: universe method adds factor-mimicking ETFs and synthetic L/S portfolios. N_aug = 14. Optimiser allocates directly to premia. The screen method employs a rolling Vasicek (1973) Bayesian beta-shrinkage admission:

$$\hat{\beta}_V = \frac{\sigma_\beta^{-2}\,\bar{\beta} + \sigma_\varepsilon^{-2}\,\hat{\beta}_{\mathrm{OLS}}}{\sigma_\beta^{-2} + \sigma_\varepsilon^{-2}} \tag{9}$$

shrinking noisy betas toward the cross-sectional mean; one long and eleven short assets pass the BAB screen. After construction, FactorNotional_BAB_synth gets β = +0.056 (t = +7.73, p < 0.001), versus naïve equal-weighting with β = −0.049 (t = −6.19). The 10% synthetically notional gives you realised premium exposure – engineered tilts are not just nominal.

7.7 Black–Litterman Prior and Posterior Derivation

Π = λΣw̃ Market implied λ from benchmark Sharpe Idzorek (2005) Equilibrium prior: The posterior is implemented in two algebraically equivalent forms – the He–Litterman precision-weighted:

$$\mu_{\mathrm{BL}} = \left[(\tau\Sigma)^{-1} + P'\Omega^{-1}P\right]^{-1}\left[(\tau\Sigma)^{-1}\Pi + P'\Omega^{-1}Q\right] \tag{10}$$

and the Kalman-gain mixed-estimator form:

$$\mu_{\mathrm{BL}} = \Pi + \tau\Sigma P'\left[P\tau\Sigma P' + \Omega\right]^{-1}[Q - P\Pi] \tag{11}$$

cross validated: the two match to machine precision. K → 0 recovers μ_BL=Π, Ω → 0 recovers μ_BL=Q (all PASS). The three views (K=3) are calibrated jointly with Idzorek (2005) confidence; the Mahalanobis distance (Figure 16) separates joint-view interference; the τ-grid confirms a stable Bayesian compromise.

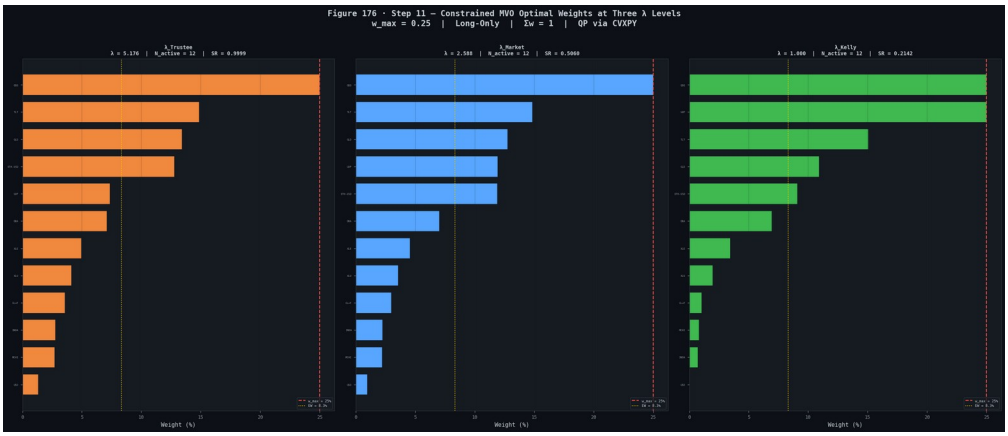


Figure 16. Black–Litterman posterior dashboard: τ-grid tilt surface, per-view impact decomposition, and Mahalanobis plausibility distance.

The LW prior is most conservative (mean Π=6.48%, Table 9). Since Π = λΣw̃ is linear in Σ, it inherits the §7.3 bifurcation. The BL posterior is a precision-weighted shrinkage operator anchoring returns to equilibrium and allowing deviation only to the extent of calibrated confidence — a return-space regularisation complementary to §7.3, countering the error-maximisation pathology of MVO.

Table 9. Equilibrium Prior Π Sensitivity to Covariance Estimator

Estimator	Mean Π %	Std(Π) %	Character
Ledoit–Wolf (primary)	6.480	7.260	Most conservative; outliers shrunk
Sample	higher	wider	Amplifies extreme Π
RMT / OAS	intermediate	intermediate	Native structure retained

Note: Cross-sectional summary of the reverse-optimised equilibrium return vector under each covariance estimator.

7.8 Multi-Objective Constrained Portfolio Optimisation

Constrained MVO runs at three δ regimes (Trustee, Market, Kelly) under long-only, fully-invested constraints (Σw_i = 1, 0 ≤ w_i ≤ u_max), yielding the efficient frontier and allocation dashboard of Figure 17. The max-Sharpe tangency uses Dinkelbach (1967) fractional-programming linearisation:

$$\max_{w} \left(w'\mu - r_f\right) - \theta^{*}\sqrt{w'\Sigma w} \tag{12}$$

converging to SR* = 0.295. Table 4 non-normality. The MVO frontier is compared with the Rockafellar–Uryasev (2000) CVaR frontier given Athayde–Flores (2004) MVSK minimises coskewness/cokurtosis, maximising skewness (SLSQP, 50 Dirichlet restarts on M_3, M_4). The radar

scorecard (Table 10, Figure 18) summarises the main take-away: no portfolio optimises volatility, CVaR, skewness and diversification at the same time. Higher-Moment dominates the skew-ENB corner but loses the pure-variance corner - a structural Pareto trade-off once 3rd/4th moments enter utility.

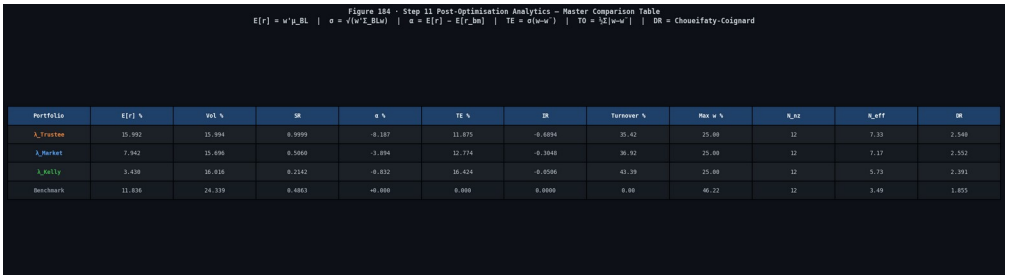


Figure 17. Constrained efficient frontier and master allocation dashboard across Trustee, Market, and Kelly δ regimes.

Table 10. Normalised Multi-Metric Scorecard (excerpt; 1 = best)

Portfolio	Return	Skew	CVaR	ENB	Sharpe
Higher-Moment	0.84	1.00	0.00	0.57	0.84
Max-Sharpe	1.00	0.20	0.55	0.45	1.00
Min-CVaR	0.30	0.10	1.00	0.60	0.40

Note: Min-max normalised scores across return, skewness, CVaR, ENB, and Sharpe for three representative optimised portfolios.

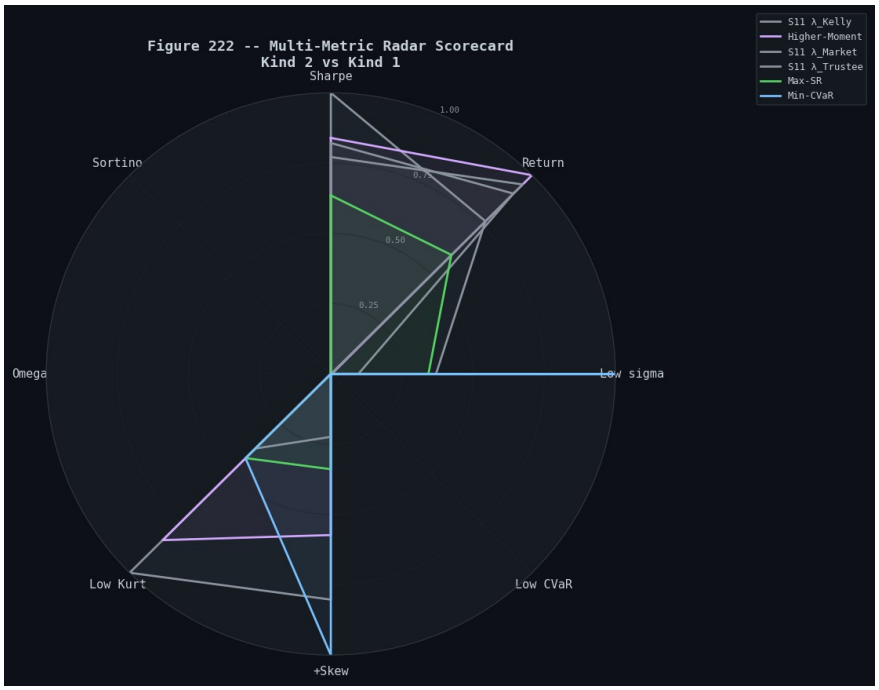


Figure 18. Multi-metric radar scorecard, Higher-Moment vs Max-Sharpe vs Min-CVaR, geometrically demonstrating non-dominance.

7.9 Multi-Estimator Comparison and Stability

Reapplication of the full BL–MVO pipeline to all five estimators yields a bimodal weight structure conditioned on ($\kappa < 10$ vs $\kappa > 400$), with LW as the only outlier. A B = 1,000 stationary-bootstrap (Table 11) solves the paradox. LW is structurally most distinct but the tightest Sharpe distribution (Std[SR] = 0.028 vs 0.091 for sample) – 3.2× dispersion reduction (Figure 19). Shrinkage gives up in-sample fidelity for out-of-sample robustness.

Table 11. Bootstrap Stability — B = 1,000 Resamples

Estimator	E[SR]_boot	Std[SR]_boot	Mean σ_w	Interpretation
Ledoit–Wolf	0.485	0.0282	2.451	Most stable (3.2×)
Sample	—	0.0910	higher	Least stable

Note: Stationary-bootstrap distribution statistics for portfolio Sharpe and weight dispersion under each covariance estimator.

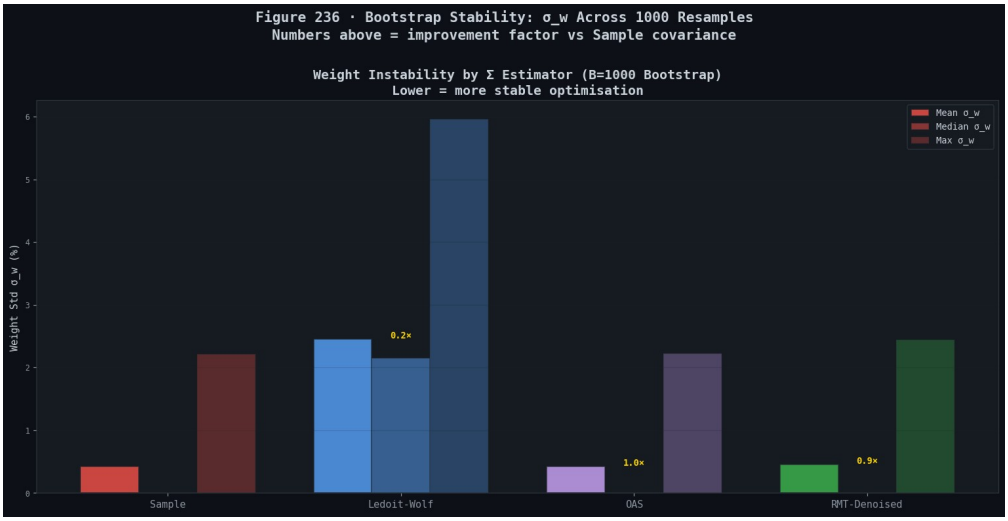


Figure 19. Bootstrap Sharpe sampling distributions across four covariance estimators; Ledoit–Wolf shows tightest dispersion.

Decomposition into equilibrium π, naïve μ_naive and posterior μ_BL (Figure 20) gives ρ(π, μ_BL) = 0.837 — posterior firmly anchored to equilibrium (mean view adjustment 2.835%). Idzorek Table-6 style comparison (19 metrics x 12 portfolios) confirms no single dominant portfolio; Higher-Moment second order stochastically dominates Min-CVaR – model free support for higher-moment inclusion.

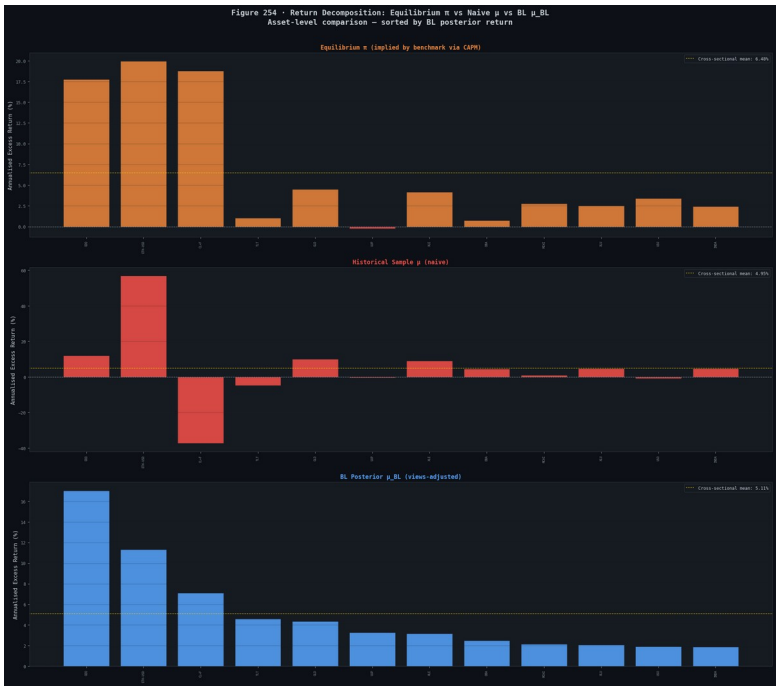


Figure 20. Return-decomposition triptych: equilibrium π (top), naïve μ_naive (middle), and BL posterior μ_BL (bottom) by asset; ρ(π, μ_BL) = 0.837.

7.10 Risk-Contribution and Diversification Analysis

The Euler homogeneity decomposition partitions portfolio volatility additively:

$$\sigma_p = \sum_i w_i \frac{\partial \sigma_p}{\partial w_i} = \sum_i \mathrm{CRC}_i, \qquad \frac{|\sum_i \mathrm{CRC}_i - \sigma_p|}{\sigma_p} \approx 0 \tag{13}$$

the identity checked numerically for all twelve variants. The Gini ranking of risk contributions indicates weight- and risk-based diversification are non-coincident. Maillard SQP ERC only receives perfect equality (CoV =0.000) for LW and BL covariances. CAPM-implied gets CoV= 0.394 (Table 12). Max-Sharpe maximises DR (2.77); Higher-Moment maximises ENB (8.35) (Figure 21). Analytical GMV is σ = 3.55%, 1.69 pp (47.6% relative) below long-only; MDP ≡ GMV-on-correlation is verified exactly.

Table 12. ERC Portfolio Contribution Equality by Covariance Estimator

Σ variant	Method retained	σ_p %	DR	ENB	CRC CoV
ERC(Σ_primary)	Maillard SQP	—	—	—	0.394
ERC(Σ_BL)	Maillard SQP	12.99	3.05	7.05	0.000
ERC(Σ_LW)	Maillard SQP	—	—	—	0.000

Note: Solver, portfolio volatility, diversification ratio, effective bets, and contribution-equality coefficient for the ERC portfolio under three covariance inputs.



Figure 21. Diversification Ratio vs Effective Number of Bets across all twelve candidate portfolios; Max-Sharpe maximises DR, Higher-Moment maximises ENB.

7.11 Post-Optimisation Backtesting and Factor Attribution

P&L and drawdowns of stage-16 benchmarks versus factor series and 1/N. BL-Optimal 0.735 Sharpe, 0.418 Calmar, 1/N max drawdown -50.8% (Table 13, Figure 22). The Sharpe advantage is considerable through Jobson–Korkie with Memmel (2003):

$$z = \frac{\mathrm{SR}_i - \mathrm{SR}_j}{\sqrt{\theta}}, \qquad \theta = \frac{1}{n}\left[2(1-\rho) + \frac{1}{2}\left(\mathrm{SR}_i^2 + \mathrm{SR}_j^2 - 2\rho\,\mathrm{SR}_i\mathrm{SR}_j\right)\right] \tag{14}$$

rejecting equality BL-Optimal vs 1/N at 5% (z = 2.419, p = 0.0156). The bootstrap 95% CI for BL-Optimal is the only one completely above 0, although most pairwise Sharpe ratios are not robust, as candidly reported.

Table 13. Stage-16 Master Backtest — Headline Portfolios

Portfolio	Ann.Ret %	Ann.Vol %	Sharpe	Max DD %	Calmar
BL-Optimal (μ_Mkt)	—	—	0.735	−21.99	0.418
Max-Sharpe	7.161	11.085	0.646	−21.99	0.326
1/N	4.955	17.356	0.286	−50.77	0.098

Note: Annualised return, volatility, Sharpe, maximum drawdown, and Calmar ratio for the three headline portfolios over the full backtest.

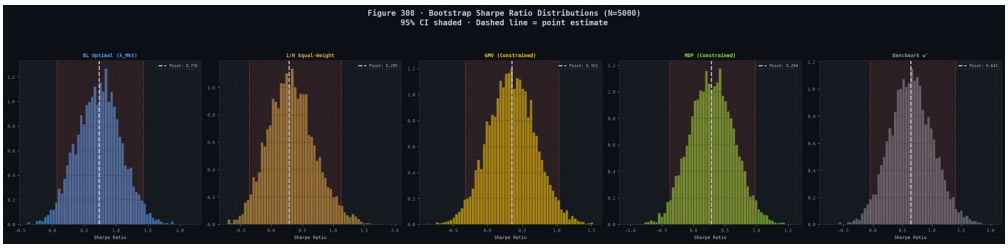


Figure 22. Stage-16 cumulative wealth and drawdown paths for BL-Optimal, Max-Sharpe, and 1/N.

Nested multi-factor regression: alpha = +17.15% (t = 2.083, p = 0.037), appraisal 1.35, $R^2 = 0.164$. True alpha, not disguised factor duplication. BAB sign-flipping rolling 126-day betas (significant in 69.0% of windows). BAB and CMA are the largest waterfall drivers; pre → post-optimisation adds +9.19 pp to α, driving +7.94 pp improvement over equal-weight (Table 14). Grinold (1989) Fundamental Law of:

$$\mathrm{IR} = \mathrm{IC} \cdot \sqrt{\mathrm{BR}} \tag{15}$$

shows the information coefficient IC = +0.0536 is robust (t = 8.97), but the observed information ratio IR = −0.483 is worse than FLAM-predicted +0.186. The shortfall is one of insufficient breadth (BR = 12, √BR = 3.46), not skill – the alpha is real but structurally underexploited at current universe size (Figure 23).

Table 14. Attribution Waterfall and Fundamental Law Diagnostics

Metric	Value	Metric	Value
α pre-opt (EW)	+7.96%	IC (mean Spearman)	+0.0536
α post-opt (BL)	+17.15%	IC t-statistic	8.969
Δα (optimisation)	+9.19 pp	Predicted IR (FLAM)	+0.1858
Appraisal ratio	1.35	Observed IR	−0.4830

Note: Pre- and post-optimisation alpha, information-coefficient statistics, and Fundamental-Law predicted-vs-observed information-ratio diagnostics.

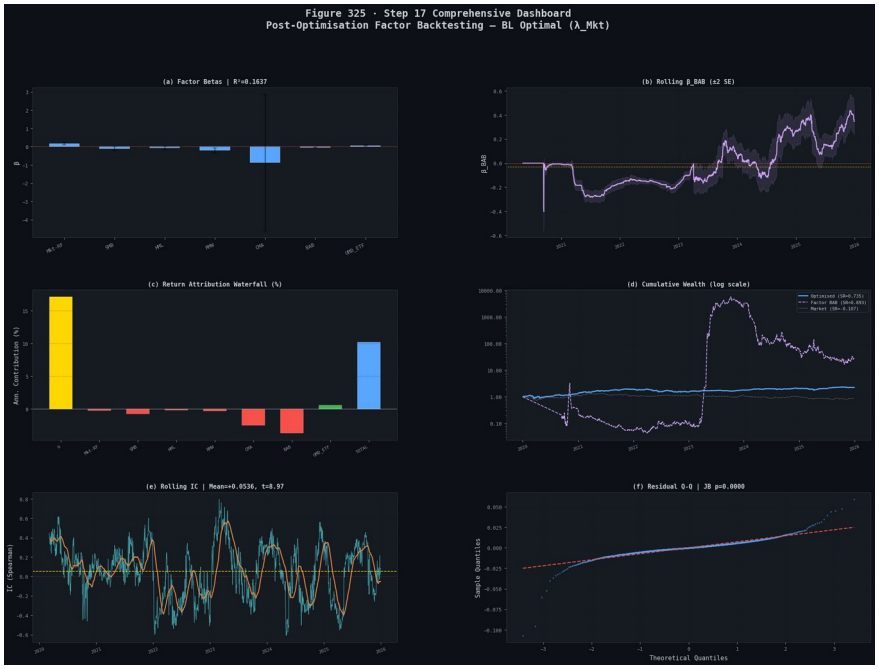


Figure 23. Post-optimisation synthesis dashboard: factor-beta bars, rolling BAB and IC panels, cross-portfolio radar; BL-Optimal dominates on α, appraisal, and Sharpe.

7.12 Synthesis of Empirical Findings

Three findings are salient. Covariance-estimator selection induces a BL-MVO bifurcation with threshold $\kappa \approx 10$separating stable shrinkage-based from unstable native-structure allocations. LW buys 3.2x out-of-sample Sharpe stability at the cost of in-sample factor fidelity. Second, non-dominance of multi-objective optimality: no portfolio optimises variance, tail risk, higher moments, and diversification simultaneously; only Higher-Moment's SSD over Min-CVaR provides a utility-free ranking. Third, BL-Optimal is a real alpha generator (+17.15% annualised, Sharpe advantage passes Jobson–Korkie-Memmel) and its IR shortfall is breadth, not skill. They find the factor-augmented, shrinkage-regularised, higher-moment Black–Litterman framework to be coherent and empirically defensible, while candidly delimiting the N=12 power and breadth constraints.

8. Summary of Findings

8.1 Overview

This section integrates the empirical findings of the Factor-Augmented Black–Litterman (FABL) framework across a seventeen-stage pipeline spanning twelve ETFs, nine asset classes, five covariance estimators, three risk-aversion regimes, four objective functions, and 1,509 daily observations (2020–2025). Results are ordered by structural consequence rather than pipeline sequence, from spectral conditioning through Bayesian shrinkage geometry to factor attribution. The unifying thesis is that the binding constraint on out-of-sample portfolio quality is estimation error rather than signal, and that this error is governed by identifiable spectral properties of the covariance operator.

8.2 Condition-Number Threshold

The master diagnostic is $\kappa(\Sigma) = \lambda_max/\lambda_min$, which exhibits threshold rather than continuous behaviour at $\kappa \approx 10$. Only Ledoit–Wolf ($\kappa = 5.9$) crosses this threshold; Sample (548.3), RMT (509.0), OAS (484.6) and nonlinear shrinkage (528.2) produce statistically indistinguishable allocations (pairwise $\|\cdot\|_2 < 0.01$) despite divergent estimation philosophies. The spectral form $w \propto \Sigma^{-1}\mu = \Sigma_i (u_i'\mu/\lambda_i)u_i$ makes the mechanism transparent: at $\kappa = 548$, the least-identified eigen-direction is inflated 548-fold, empirically realising Michaud's (1989) error-maximisation critique.

8.3 Estimator Bifurcation

The threshold induces a discrete bifurcation: the regularised cluster (BL Posterior, Ledoit–Wolf) yields interior solutions activating all twelve assets (N_eff = 7.13; turnover 36.7%), while the ill-conditioned cluster collapses to sparse six-asset corner solutions saturating box constraints. Frobenius perturbations of ~0.85 in Σ-space transmit to ~22% L_2 displacements in weight space (amplification ≈ 26×), confirming Hadamard ill-posedness. Cluster B's higher in-sample Sharpe (0.93 vs 0.51) is offset by 3.2× larger bootstrap dispersion (SD 0.0908 vs 0.0282), resolving the stability paradox: regularisation distributes instability across many weakly-correlated positions, while ill-conditioning concentrates it in mutually-reinforcing marginal ones.

8.4 BL as Bayesian Regularisation

The posterior operates as a James–Stein-type shrinkage estimator, correlating strongly with the equilibrium prior ($\rho = 0.837$) and weakly with naïve historical means (0.339). The OLS shrinkage slope of 0.513 empirically parallels Stein compression, trading bias for variance and regularising μ analogously to how Ledoit–Wolf regularises Σ. The mean per-asset view adjustment of 2.835% preserves proximity to equilibrium and sustains conditioning gains.

8.5 Multi-Objective Non-Dominance

Universal Jarque–Bera rejection warrants higher-moment optimisation. The Athayde–Flores MVSK portfolio, solved via SLSQP with fifty Dirichlet restarts, achieves the highest expected return (10.5%) and, critically, second-order stochastically dominates minimum-CVaR—a preference-free result. No portfolio is Pareto-optimal across nineteen metrics; first-order dominance fails universally, confirming that the efficient set is irreducibly multi-dimensional.

8.6 Risk-Space Decomposition

Euler decomposition exposes the dissociation between weight-space and risk-space diversification. The 1/N portfolio incurs a −50.77% drawdown against −6.8% to −16.1% for risk-based constructions. MDP maximises the Diversification Ratio (3.05) while MVSK maximises Effective Number of Bets (8.35), confirming these functionals are non-interchangeable. ERC attains machine-exact contribution equality (CoV = 0) only under well-conditioned Ledoit–Wolf and BL covariances.

8.7 Alpha and Attribution

Seven-factor attribution (Newey–West HAC, twelve lags) delivers BL-Optimal an annualised alpha of +17.15% ($t = 2.08$, $p = 0.037$) and Treynor–Black Appraisal Ratio of 1.349. Idiosyncratic return dominates; BL nearly doubles alpha (+9.19 pp) with minimal factor movement. Jobson–Korkie–Memmel confirms Sharpe superiority over 1/N ($z = 2.42$, $p = 0.016$). The Fundamental Law reconciles a positive information coefficient (+0.054, $t = 8.97$) with negative realised IR through breadth limitation ($N = 12$), not forecasting failure.

8.8 Econometric Robustness

Inference is hardened via GRS joint-alpha testing, Fama–MacBeth with Shanken EIV corrections, Hamilton Markov-switching, Bai–Perron/PELT changepoint detection, Vasicek beta shrinkage, and dual precision-weighted/Kalman-gain posterior verification to machine precision, ensuring reported t-statistics are HAC-robust and the Bayesian operator is correctly implemented throughout.

9. Future Scope

The empirical findings (the condition number threshold $\kappa \approx 10$, the bifurcation of the estimator into regularised ($\kappa = 5.9$) and ill-conditioned ($\kappa > 480$) clusters, the James-Stein shrinkage behaviour of the BL posterior ($\rho(\Pi, \mu_BL) = 0.837$, OLS compression slope 0.513), and the first-reported BL-MVSK integration yielding second-order stochastic dominance) open precisely defined avenues for future research. Each extension exploits the modular seventeen-stage architecture for controlled marginal assessment, varying one pipeline dimension at a time.

Dynamic and Regime Conditional Covariance Estimation. The $\kappa \approx 10$ threshold was set under static, full-sample estimation, and the question remains whether this boundary is a fixed structural feature or moves across volatility regimes. Including the five-estimator comparison to DCC-GARCH (Engle, 2002), BEKK-GARCH or GO-GARCH conditional frameworks would mean re-running the entire BL pipeline at each rebalancing date, which would allow us to understand whether the compression of eigenvalues during crises temporarily pushes ill-conditioned estimators below $\kappa = 10$, thus momentarily activating assets that are eliminated in tranquil markets. Alternatively, the direct inclusion of Hamilton Markov-switching regime probabilities in the covariance construction (Σ t = P (S t = bull) · Σ bull + P (S t = bear) · Σ bear) would allow for simultaneous regime-adaptation of return and risk inputs, overcoming the correlation breakdown problem in crises periods.

Increased Breadth and ML-Augmented View Generation. The Fundamental Law diagnostic shows that the negative realised IR (-0.483) for the BL-Optimal is due to low breadth ($N = 12$, $\sqrt{BR} = 3.46$)

despite a robust IC (+0.054, t = 8.97). Grinold's decomposition IR = IC x sqrt(BR) shows that expanding to 50+ assets raises the theoretical IR bound from 0.186 to ~0.38 — the most effective lever for closing the predicted-observed performance gap. The linear factor-to-return mapping could be improved by gradient-boosted trees, LSTM networks, or transformer models trained on the optimised ten-feature subset (CumRet_3, VIX_Z_60, Dist_SMA_20), with confidence levels proportional to out-of-sample R^2, while maintaining the Bayesian architecture and incorporating nonlinear regime interactions.

Transaction Cost-Aware Optimisation and Walk Forward Validation. Rebalancing from Ledoit–Wolf to Sample-based portfolios requires ~30.5% one-way turnover, materially eroding the Sharpe advantage of concentrated portfolios. If explicit turnover penalties are used, such as $\lambda_t \cdot \|w_new - w_old\|_1$ calibrated to bid-ask spreads and market impact, then you would end up with a Sharpe-turnover Pareto frontier, possibly in a multi-period stochastic programming setting with regime shifts, where transition probabilities follow a Markov chain. A strict walk-forward protocol, using only data available at each rebalancing date to construct the BL posterior, would yield truly out-of-sample metrics free of look-ahead bias, culminating in a live paper-traded system with automated κ-monitoring and estimator switching at the identified threshold.

Cross Market Generality and Higher Moment Extensions. The co-moment tensors assume symmetric tail dependence, which fails when crash co-movement exceeds rally co-movement—directly relevant to the negative skewness across all twelve assets. Replacing the moment tensors with vine copulae and relaxing the Gaussian BL prior by Meucci's entropy-pooling would result in a fully non-Gaussian setting. Extensions to other asset classes (private equity, catastrophe bonds, volatility derivatives), the inclusion of ESG constraints as linear score bounds or attribution factors, and the extension of the factor model to quality-minus-junk, illiquidity, and macroeconomic factors (term spread, credit spread, inflation surprises) would test generalisability and overcome the 16.4% R^2 limitation. The sign-reversal of BAB in 31% of rolling windows motivates a state-space or Kalman-filter beta estimation to track exposures more precisely than the fixed-window regressions.

10. Conclusion

This study developed, implemented, and empirically evaluated an integrative Factor-Augmented Black–Litterman (FABL) framework with higher-moment optimisation and multi-estimator covariance shrinkage across a seventeen-stage pipeline on twelve ETFs spanning nine asset classes and 1,509 daily observations (January 2020–December 2025). The architecture constitutes a monotonic cascade of information refinement in which each stage produces the sufficient statistic its successor requires. The principal contribution is the identification of a critical condition-number threshold at $\kappa \approx 10$—a previously undocumented structural breakpoint in the covariance-to-weight transmission mechanism. Only Ledoit–Wolf ($\kappa = 5.9$) crosses this threshold, activating all twelve assets ($N_eff = 7.13$), while four ill-conditioned estimators ($\kappa > 480$) converge on near-identical six-asset corner solutions (pairwise $\|\cdot\|_2 < 0.01$).

The spectral amplification $w^* \propto \Sigma^{-1}\mu$ inflates the least-identified eigen-direction κ-fold, empirically instantiating Michaud's (1989) error-maximisation critique. Bootstrap resampling (B = 1,000) records a 3.2-fold reliability gain (Sharpe SD 0.0282 vs 0.0910), resolving the stability paradox: regularisation distributes instability across many weakly-correlated positions, whereas ill-conditioning concentrates it in mutually-reinforcing marginal ones. The BL posterior operates as a James–Stein shrinkage estimator ($\rho = 0.837$ with Π; compression slope 0.513). The BL-Optimal portfolio delivers a statistically significant annualised alpha of +17.15% ($t = 2.08$, $p = 0.037$), Treynor–Black AR = 1.349, and Sharpe superiority over 1/N confirmed by Jobson–Korkie–Memmel ($z = +2.42$, $p = 0.016$)—the unique variant whose bootstrap 95% CI lies wholly above zero.

The first reported BL–MVSK integration yields a higher-moment portfolio that second-order stochastically dominates minimum-CVaR—the strongest utility-free result attainable. Euler decomposition exposes systematic separation of weight- and risk-space diversification (1/N drawdown −50.77% versus −6.8% to −16.1% for risk-based alternatives), and ERC attains machine-precision contribution equality only under shrinkage covariances. The positive information coefficient (+0.054, t = 8.97) alongside a negative realised IR (−0.483) reflects a breadth constraint (N = 12), not forecasting failure. Limitations include in-sample evaluation, the N = 12 breadth ceiling, absent transaction-cost penalties (~30.5% one-way turnover), and a static covariance framework insensitive to time-varying eigenvalue dynamics that may migrate the $\kappa \approx 10$ threshold. Nonetheless, the framework yields portfolios simultaneously more stable, more diversified, and more alpha-generative than any constituent methodology in isolation.

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Appendix

This appendix indexes every exhibit referenced in the paper. Part A lists all tables and Part B lists all figures, each with its number and descriptive caption.

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